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Design and Coordinated Motion Control of an Automated Unmanned Aerial Vehicle Docking System for Unattended Industrial Operations

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Abstract: Automated unmanned aerial vehicle (UAV) docking systems play an important role in unattended industrial inspection and monitoring. However, coordinating mechanical movements, maintaining landing-platform stability, and securing aircraft during deployment and recovery remain challenging. This study investigates the mechanical design and coordinated motion control of a four-position UAV docking system. An integrated architecture comprising a sliding roof, a three-degree-of-freedom (3-DOF) landing platform, and a mechanical centering and locking mechanism was developed. The roof and centering drives were analysed through mechanical load calculations and parameter sensitivity analysis, while a supervisory control framework was formulated to coordinate actuator movements through position feedback and operating interlocks. A representative three-actuator model was also established to evaluate platform levelling under static inclination and periodic disturbances using proportional (P) and proportional-derivative (PD) control. The calculated roof resistance and driving torque were 250 N and 6.75 $\mathrm{N}\cdot\mathrm{m}$, respectively, while the centering resistance was 325 N and the estimated motor torque was 4.51 $\mathrm{N}\cdot\mathrm{m}$ under an assumed unit transmission ratio. In the simplified numerical model, P and PD control achieved static settling times of 1.02 s and 1.21 s, respectively. Under periodic disturbances, the combined root-mean-square (RMS) attitude error decreased from 2.599° without compensation to 1.299° with P control and 1.185° with PD control. The results indicate that coordinated mechanical operation and feedback-based platform levelling can be integrated within a common docking framework. The study provides a basis for the development of unattended industrial UAV docking systems, although the predicted control performance requires validation using the actual platform geometry and actuator characteristics.

Keywords: Automated unmanned aerial vehicle docking system; Unattended industrial operations; Coordinated motion control; Three-degree-of-freedom landing platform; Platform levelling; Mechanical centering and locking

1 Introduction

Unmanned aerial vehicles (UAVs) are increasingly used for industrial inspection, infrastructure monitoring, emergency response, and operations at geographically dispersed sites. Their ability to collect information without direct human access is particularly useful for power transmission networks, industrial facilities, and remote installations. However, unattended UAV operation requires more than autonomous flight. Aircraft must also be recovered, positioned, secured, serviced, and prepared for subsequent missions with limited human intervention. Automated docking stations provide the mechanical and operational infrastructure needed to support these activities. Grlj et al. [1] reviewed the development of fixed and mobile UAV docking stations, including landing platforms, aircraft storage, and energy replenishment. More recently, Šćuric et al. [2] examined automated multirotor docking and charging systems, with particular attention to positioning mechanisms, landing arrangements, and battery-replacement technologies. Galimov et al. [3] classified active and passive UAV positioning mechanisms used in landing stations and discussed their engineering characteristics. These studies demonstrate that the reliability of an automated docking facility depends not only on aircraft landing accuracy but also on the mechanical functions performed after touchdown.

The design of unattended docking equipment has attracted increasing attention as UAV applications have expanded beyond individual flights. Stuhne et al. [4] investigated the design and validation of a wireless drone docking station, providing an example of the integration of aircraft recovery with station-side equipment. Muñoz-Gómez et al. [5] developed an autonomous charging arrangement for UAV operations on overhead power lines, demonstrating a different approach to supporting repeated industrial inspection missions. Zhang et al. [6] developed a shipborne UAV recovery system integrating an arresting mechanism, compliant robotic grasping, and automated aircraft storage. Although these systems differ in configuration and operating environment, they illustrate the importance of mechanical positioning, service interfaces, and reliable aircraft recovery. For a docking station accommodating several UAVs, these requirements become more demanding because aircraft storage, internal transportation, launching, and recovery must be performed within a shared mechanical structure.

Autonomous docking has also been investigated in other industrial robotic systems. Jia et al. [7] developed a vision- and lidar-based docking and recharging method for a mobile robot operating in a manufacturing environment. Feng et al. [8] examined autonomous alignment and docking control for a self-reconfigurable modular robotic system, considering the coordination of positioning and mechanical engagement. Although these systems do not employ the same mechanical configuration as a UAV docking station, they demonstrate the importance of combining position feedback, mechanical alignment, and controlled docking operations. Similar requirements arise when a UAV must be repositioned after landing and mechanically secured before transfer into storage. In such cases, the docking mechanism must compensate for residual positioning errors without interfering with the aircraft structure or surrounding equipment.

Autonomous landing on moving platforms represents another important research direction. Cheng et al. [9] investigated UAV–Unmanned Ground Vehicle (UGV) autonomous docking in Global Positioning System (GPS)-denied environments using relative positioning and multisensor information. Keipour et al. [10] developed a visual servoing approach for UAV landing on a moving vehicle without relying on external motion-capture equipment. Guo et al. [11] examined robust visual landing control under delayed measurements and external disturbances. Aikins et al. [12] investigated moving-platform landing under partial observability using a recurrent reinforcement-learning approach. More recently, Byun et al. [13] proposed a vector-field-based landing strategy that considered the detectability of a moving target, while Wu et al. [14] examined high-precision visual landing using object detection. These studies have advanced aircraft-side guidance and relative localization. Nevertheless, successful aircraft guidance does not guarantee that the receiving platform is correctly oriented or that the aircraft can be mechanically restrained after touchdown.

The recovery problem becomes more complicated when the supporting structure is subjected to continuous motion. Li et al. [15] investigated coordinated UAV–Unmanned Surface Vehicle (USV) approaching and landing, incorporating predictive control and feedback regulation during different landing stages. Tian et al. [16] examined a UAV–USV docking system based on deck-motion compensation and attitude prediction. Morales et al. [17] studied vision-based autonomous following and landing on a moving platform. These investigations indicate that support motion, sensing uncertainty, and relative positioning influence the landing process. However, aircraft-side compensation and station-side platform adjustment address different aspects of the same recovery problem. The former primarily regulates aircraft movement relative to the target, whereas the latter changes the mechanical orientation of the receiving surface. An unattended docking station installed on an inclined or moving support may therefore require an adjustable landing platform in addition to the aircraft guidance system.

Parallel motion platforms provide a useful basis for examining adjustable landing-surface mechanisms. Zhao et al. [18] investigated the kinematics and statics of the Gough–Stewart platform, demonstrating a systematic relationship between platform motion, actuator geometry, and mechanical loading. Silva et al. [19] examined Stewart-platform motion control using industrial controllers, including kinematic calculations and trajectory execution. These studies show that the required actuator movements depend on the geometric configuration of the platform and its intended motion. For a three-degree-of-freedom (3-DOF) landing platform, the relevant controlled motions may include vertical displacement, roll, and pitch. The corresponding actuator displacements must be calculated from the desired platform pose while respecting the available stroke and mechanical joint constraints. Although the present docking system employs a different mechanism from a conventional six-degree-of-freedom Stewart platform, related kinematic methods provide a useful foundation for preliminary motion analysis.

The coordination of individual actuators is another important consideration. Zhou et al. [20] investigated synchronization control with dynamics compensation for a three-axis parallel motion platform, considering the interaction between motor movements and the resulting tracking performance. Lv et al. [21] developed an anti-swaying control strategy for a ship-mounted three-revolute–cylindrical–universal (3-RCU) parallel platform using dynamic gravity compensation. Li et al. [22] examined a neural-network-assisted proportional–integral–derivative (PID) control method for wave compensation using a shipborne Stewart platform. Tang et al. [23] investigated a control method for a swing-type wave-compensation platform incorporating feedback regulation and mechanical motion constraints. These studies addressed different platform structures and control objectives, but they demonstrate

that motion accuracy depends on both the feedback strategy and the dynamic characteristics of the supporting mechanism. In a three-actuator landing platform, coordinated displacement commands are required because the movement of each actuator contributes to the resulting platform attitude.

The mechanical configuration and structural characteristics of parallel platforms also affect their attainable performance. Fang et al. [24] investigated an active vibration-isolation system based on a piezoelectric Stewart platform. Zhang et al. [25] examined the additional stiffness introduced by flexible joints in a Stewart mechanism, while Tao et al. [26] studied multi-objective optimization of Stewart-platform structural parameters. These investigations illustrate the influence of joint characteristics, structural parameters, and workspace limitations on platform behaviour. Such factors are relevant when evaluating an adjustable UAV landing platform, particularly under inclination disturbances or repeated operating cycles. A simplified kinematic representation may be sufficient for preliminary control analysis, but its predicted response must be distinguished from the behaviour of the manufactured mechanism, which depends on the actual attachment coordinates, joint arrangement, actuator force capacity, and structural stiffness.

Control methods for closed-chain mechanisms provide additional references for actuator coordination. Nguyen et al. [27] investigated decentralized adaptive control for manipulators based on closed kinematic chains, addressing synchronized motion under model uncertainty and load variation. Their work differs from the present application in both mechanism configuration and control complexity, but it illustrates the importance of maintaining coordinated motion when several actuators determine a common platform pose. For industrial docking equipment, a control strategy must also respect actuator stroke and velocity constraints. A controller that reduces attitude error under nominal conditions may not remain feasible when large inclination disturbances require actuator movements beyond the available mechanical range. Consequently, platform levelling should be evaluated using both attitude-error measures and actuator-motion requirements.

Recent aircraft-side control studies further emphasize the influence of dynamic disturbances on UAV recovery. Cho et al. [28] investigated autonomous ship-deck landing using feed-forward image-based visual servoing. Yang et al. [29] developed an online predictive visual servoing approach that considered target visibility and disturbance effects during rotor-UAV landing. Wu et al. [30] examined sliding-mode control for vision-based high-precision UAV landing under disturbances. These studies provide relevant evidence of the challenges associated with landing on moving supports. However, their principal emphasis remains on aircraft guidance and control. The mechanical operations required after touchdown, including centering, locking, lifting, and transfer into storage, introduce additional constraints that must be addressed by the docking station itself.

For a multi-position UAV docking station, the operation of individual mechanisms cannot be considered independently. A protective roof must provide sufficient clearance before the lifting platform enters the external operating region. The transfer mechanism must establish mechanical engagement before a parking platform is released from its storage position. Following recovery, the centering mechanism must correct the aircraft position, and the locking elements must secure the landing gear before internal transportation. The adjustable landing platform must also reach its required position and attitude before the system authorizes take-off or landing. These requirements introduce a coordination problem involving continuous actuator motion and discrete operating states. Reliable operation therefore depends on both the mechanical capabilities of individual subsystems and the supervisory conditions governing their interaction.

Despite the progress reported in the literature, the available studies generally address separate aspects of automated UAV recovery. Docking-station research has examined positioning arrangements, charging equipment, and recovery mechanisms [1–8]. Aircraft-side studies have concentrated on relative localization, landing guidance, and disturbance rejection [9–17, 28–30]. Research on parallel platforms has investigated kinematics, structural characteristics, actuator synchronization, and motion compensation [18–27]. These contributions provide useful foundations, but they do not individually resolve the integration requirements of a four-position UAV docking station incorporating a protective roof, shared landing platform, vertical lifting and horizontal transfer mechanisms, and mechanical centering and locking. In particular, mechanical loading, actuator limitations, platform attitude regulation, and permissible operating sequences need to be considered within a consistent engineering framework.

The system investigated in this paper is a four-position automated UAV docking station intended for unattended industrial deployment and recovery. Four aircraft are accommodated in storage positions arranged around a central transfer and landing area. During deployment, the selected aircraft is transferred from storage to the central platform, raised to the operating position, and prepared for take-off. Recovery follows a corresponding sequence, with additional operations for platform positioning, aircraft centering, mechanical locking, and transfer into storage. The station incorporates a sliding protective roof driven by a rack-and-pinion mechanism, 3-DOF landing platform equipped with servo-driven electric actuators, and a lead-screw-based centering assembly. Attitude feedback and inverse kinematics are considered for platform levelling, while position sensing and interlocking conditions govern the sequence of mechanical operations.

The objective of this study is to investigate the mechanical design and coordinated motion control of these

subsystems, with particular attention to the relationship between actuator requirements, platform attitude regulation, and unattended operating procedures. Rather than developing a new aircraft guidance algorithm, the work addresses the docking equipment that supports aircraft deployment and recovery. The available engineering parameters were used to assess the roof and centering drives through analytical calculations and parameter sensitivity analysis. A supervisory control framework was formulated to describe the operating dependencies between the mechanisms. In addition, a representative three-actuator model was established to examine closed-loop platform levelling under static inclination and periodic disturbance conditions using proportional (P) and proportional-derivative (PD) feedback. The numerical evaluation employed assumed geometric and actuator parameters and was not intended to represent experimental validation of the manufactured docking platform.

The principal contributions of this study are threefold. First, an integrated operating architecture is developed for a four-position UAV docking station, linking roof movement, lifting, internal transfer, platform levelling, aircraft centering, and mechanical locking through defined operating states and supervisory interlocks. Second, the mechanical loads and drive requirements of the sliding roof and centering assembly are examined through analytical calculations and parameter sensitivity analysis, with particular attention to friction resistance, transmission efficiency, and operating constraints. Third, a coordinated platform-control formulation is evaluated using a representative three-actuator model, allowing P and PD feedback to be compared in terms of static settling time, disturbance-related attitude error, and actuator movement constraints. Together, these contributions provide an engineering basis for assessing mechanical design and coordinated control requirements in unattended UAV docking equipment while identifying the further verification required for physical implementation.

The remainder of the paper is organized as follows. Section 2 describes the system architecture and operating principles of the four-position UAV docking station. Section 3 presents the mechanical design and mathematical modelling of the principal mechanisms. Section 4 develops the coordinated motion-control and supervisory interlocking framework. Section 5 presents numerical calculations, sensitivity analyses, and closed-loop performance evaluation. Section 6 discusses the engineering implications and limitations, and Section 7 summarizes the main findings.

2 System Architecture and Operating Principles

2.1 Overall Configuration of the Unmanned Aerial Vehicle Docking System

The automated docking system investigated in this study was designed to accommodate four multirotor UAVs within a single enclosure. Its mechanical configuration integrates aircraft storage, horizontal transfer, vertical lifting, platform levelling, and mechanical positioning into a common operating system. Unlike a conventional single-position docking station, the proposed arrangement uses a shared central landing platform to serve multiple storage positions. This configuration reduces the need for separate launching and recovery mechanisms at each storage location, although it also requires the movements of several mechanical subsystems to be coordinated during aircraft deployment and recovery.

The overall arrangement follows a compact four-position layout. Four storage compartments are distributed on both sides of the central transfer area, with two vertically arranged storage levels. Each UAV is supported by a dedicated parking platform during storage and internal transportation. The central transfer mechanism moves the selected parking platform between its storage compartment and the common lifting and landing area. This arrangement allows individual aircraft to be deployed or recovered without requiring all four storage positions to operate simultaneously.

The principal mechanical components comprise a sliding roof assembly, a vertical lifting mechanism, a horizontal transfer mechanism, 3-DOF landing platform, and a mechanical centering and locking assembly. The sliding roof provides access to the external environment during take-off and landing while protecting the equipment during standby. The lifting mechanism transfers the parking platform between the storage levels and the external operating position. The 3-DOF platform adjusts the orientation of the landing surface, whereas the centering mechanism corrects the aircraft position after landing and secures it before storage.

The enclosure also accommodates power supply and replenishment equipment, communication devices, environmental monitoring sensors, video surveillance equipment, and a fire protection system. These auxiliary components are primarily installed in the lower section of the docking station, while environmental sensing devices are arranged around the external structure. Electrical connections and service lines are routed to the corresponding operating positions. The control system exchanges signals with the UAV and the mechanical actuators to coordinate the docking process.

The original design incorporates a centralized supervisory control arrangement. Commands are issued through the upper-level control program, while position sensors, encoders, and other feedback devices provide information about the operating conditions of the mechanisms. The movement of each subsystem is permitted only after the relevant positioning or locking conditions have been confirmed. This arrangement is particularly important for a

shared-platform system because an incomplete movement of one mechanism may obstruct the subsequent operation of another.

The overall configuration of the four-position UAV docking system is illustrated in Figure 1, showing the storage compartments, central transfer mechanism, lifting assembly, adjustable landing platform, and associated equipment. This arrangement provides the mechanical basis for the automated deployment and recovery procedures described below.

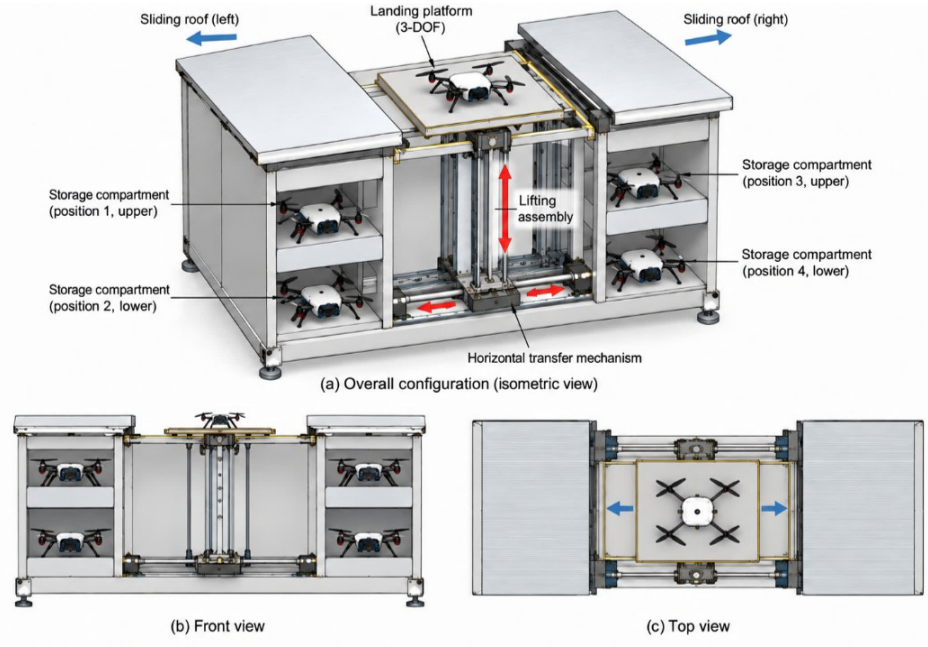


Figure 1. Overall configuration of the four-position automated unmanned aerial vehicle docking system

Note: DOF = degree of freedom.

2.2 Functional Organization and Control Architecture

The docking process involves several mechanisms with different motion characteristics and operating requirements. The sliding roof performs horizontal opening and closing movements, the lifting mechanism provides vertical transportation, and the transfer mechanism moves the parking platform between the central operating area and the selected storage compartment. The 3-DOF platform provides attitude adjustment at the landing position, while the centering and locking assembly performs the final mechanical positioning of the UAV.

Although these mechanisms operate through separate drive arrangements, their movements are not independent. For example, the lifting platform cannot safely pass through the roof opening until sufficient clearance has been established. Similarly, the horizontal transfer mechanism must not release a parking platform before the receiving structure has been positioned and the corresponding locking mechanism has engaged. During recovery, the centering and locking operations must be completed before the aircraft is transferred into storage.

The functional control architecture can therefore be divided into three interconnected levels.

At the supervisory level, the docking controller receives deployment or recovery commands, determines the selected storage compartment, and manages the overall operating sequence. It also exchanges permission and status signals with the UAV. At the mechanism-control level, individual drive units execute the commanded movements of the roof, lifting mechanism, horizontal transfer system, landing platform, and centering assembly. At the sensing and protection level, position switches, encoders, attitude sensors, and electrical protection devices provide feedback for motion completion, operating limits, and abnormal conditions. Figure 2 illustrates the functional relationships between the supervisory controller, individual mechanism controllers, and sensing and protection devices. The architecture separates operating-sequence management from actuator-level motion control while retaining the feedback and interlocking signals required for coordinated operation.

The supervisory controller uses the feedback signals to determine whether the system can proceed to the next operation. In the original design, the main operating procedure follows a sequential logic: a subsequent operation is initiated after the preceding movement has reached its required condition. Certain movements, such as roof opening and platform lifting, may overlap where the mechanical arrangement permits, but their interaction remains subject to clearance and positioning constraints.

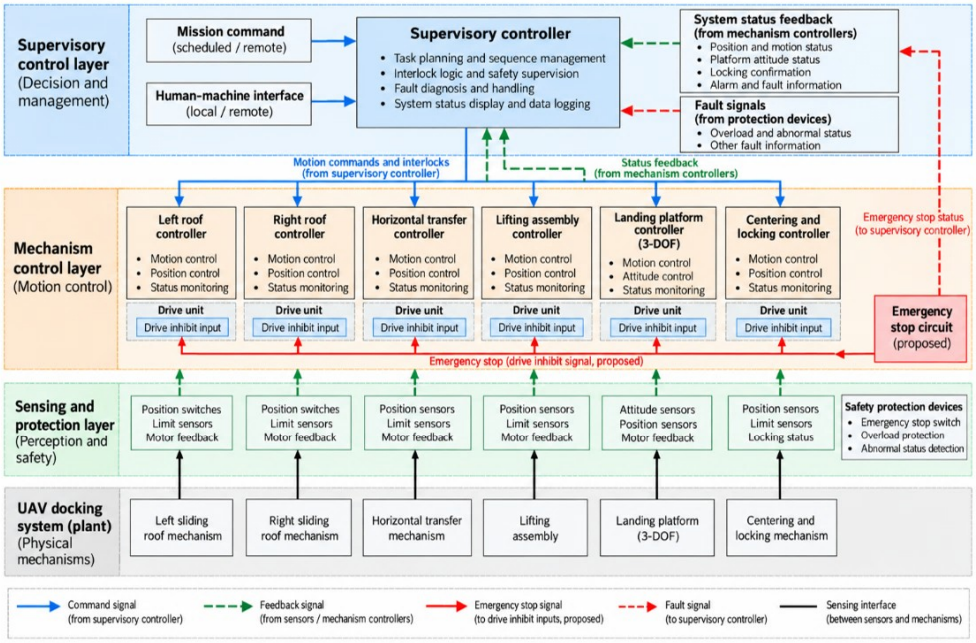


Figure 2. Functional architecture of the supervisory and motion-control system

Note: UAV = unmanned aerial vehicle; DOF = degree of freedom.

For analytical purposes, the overall docking process can be represented by a discrete operating state:

$$S(t) = \{q_r, q_l, q_t, q_p, q_c, q_k\} \tag{1}$$

where, q_r, q_l, q_t, q_p, q_c, and q_k denote the operating states of the roof, lifting mechanism, transfer mechanism, attitude-adjustment platform, centering mechanism, and locking system, respectively.

Each state variable represents a finite set of operating conditions. For instance, the roof may be closed, opening, fully open, or closing, while the lifting mechanism may be positioned at a storage level, moving, or located at the external operating position. The platform state includes the adjustment and completion conditions associated with attitude levelling.

This representation is introduced to formalize the operating sequence described in the original engineering design. It does not imply that the physical system has already implemented a particular finite-state-machine controller.

A commanded transition between operating states is permitted when the corresponding interlocking conditions are satisfied:

$$S_{j+1} = F\left(S_j, u_j, \boldsymbol{\sigma}_j\right) \tag{2}$$

where, S_j is the current operating state, u_j represents the supervisory command, and $\boldsymbol{\sigma}_j$ is the vector of sensor and interlock signals. The transition function F describes the permitted progression of the docking sequence.

For a representative operation, the authorization condition may be expressed as

$$E_j = \bigwedge_{i=1}^{n_j} \sigma_{j,i} \tag{3}$$

where, E_j is the authorization signal for operation j, $\sigma_{j,i}$ represents an individual required condition, and n_j is the number of conditions associated with that operation. The logical conjunction indicates that all mandatory conditions must be satisfied before the operation is authorized.

This formulation provides a common description of the coordination requirements without prescribing a specific controller implementation. The actual sensor assignments, timeout thresholds, and recovery procedures depend on the detailed control hardware and must be verified during system integration.

2.3 Automated Unmanned Aerial Vehicle Deployment Procedure

The deployment procedure begins when the supervisory controller receives a single-aircraft launch command. The controller first determines the required storage position and checks whether the corresponding compartment

contains an available UAV. If the selected compartment is empty, another storage position is examined. Once an available aircraft has been identified, the system initiates the preparation sequence.

During preparation, the UAV is activated and the external power and replenishment connections are disengaged. The electrical connector is withdrawn by the corresponding actuator, and the replenishment interface is disconnected. Position feedback is used to confirm that the connectors have been released. If the disconnection process exceeds the permitted time or the required confirmation signal is not received, the operating sequence is interrupted and an alarm is generated.

After the service connections have been released, the lifting mechanism moves to the selected storage level. The horizontal transfer mechanism then extends toward the corresponding compartment. When the transfer mechanism reaches its designated position, its locking assembly engages with the parking platform. The locking connection between the parking platform and the storage compartment is subsequently released. This order of operations is necessary to maintain mechanical support during the transfer of the parking platform. Releasing the storage lock before the transfer mechanism has established a secure connection could leave the platform inadequately supported. After the locking transfer has been confirmed, the horizontal mechanism retracts and brings the parking platform into the central lifting area.

The lifting mechanism then raises the parking platform toward the external operating position. The roof-opening mechanism is activated during the lifting sequence, subject to the clearance requirements between the moving platform and the roof assembly. The roof position is monitored to ensure that the opening movement is completed before the lifting platform enters the region requiring full clearance. If the roof does not reach its required position within the specified period, the lifting operation is stopped and the system enters an alarm condition.

When the parking platform reaches the designated take-off height, the vertical lifting movement is stopped. The 3-DOF landing platform then performs attitude adjustment. The attitude sensor measures the inclination of the platform, and the controller determines the corresponding actuator movements using the inverse kinematic relationship of the platform mechanism. The servo-driven electric actuators adjust the platform orientation until the required levelling condition is reached.

The original operating procedure also includes a centering operation before launch. This operation is intended to establish the required aircraft position on the parking platform. The centering mechanism is driven to its designated position, and encoder or position feedback is used to confirm completion. After centering has been completed, the aircraft restraints must be released before take-off. The controller should confirm both the required aircraft position and the release of the locking elements before issuing launch authorization. After the positioning and launch preparation conditions have been satisfied, the docking controller exchanges permission signals with the UAV. The aircraft is permitted to take off only after the mechanical system has reached the required operating state.

Following confirmation that the UAV has departed, the attitude-adjustment platform returns to its reference position. The lifting mechanism then moves the empty parking platform toward the selected storage level. The roof closes when the mechanical clearance conditions permit. The horizontal transfer mechanism returns the parking platform to its storage compartment, where the storage locking assembly engages before the transfer mechanism releases its connection.

The deployment procedure is completed when the moving mechanisms have returned to their designated standby positions and the relevant locking conditions have been confirmed.

2.4 Automated Unmanned Aerial Vehicle Recovery Procedure

The recovery procedure follows a related operating sequence but introduces additional requirements associated with aircraft landing, mechanical centering, and secure storage.

After receiving a single-aircraft recovery command, the supervisory controller identifies an available storage compartment. The compartment occupancy signals are checked to prevent the recovered aircraft from being transferred into an occupied position. Once a suitable compartment has been selected, the lifting and horizontal transfer mechanisms are positioned to retrieve the corresponding parking platform.

The horizontal transfer mechanism moves toward the selected compartment and establishes a mechanical connection with the parking platform. The storage locking mechanism is then released, allowing the platform to be transferred into the central lifting area. The lifting mechanism subsequently raises the platform toward the external landing position while the roof-opening sequence is executed.

After the platform reaches the required operating height, the 3-DOF attitude-adjustment mechanism is activated. The platform inclination is measured, and the electric actuators adjust the landing surface to the required orientation. The docking controller then communicates with the UAV to establish the landing permission condition.

Once the aircraft has landed, the mechanical centering assembly is activated. The original design uses transverse and longitudinal centering bars arranged at different heights to avoid interference during movement. The bars move toward the central position and correct the aircraft location on the parking surface. The longitudinal bars primarily

provide contact with the landing gear, while the transverse assembly incorporates locking elements that engage with the corresponding structural members of the aircraft.

The centering movement is produced through a trapezoidal lead-screw transmission. A central drive arrangement transmits motion to the corresponding centering bars, allowing the aircraft to be repositioned and restrained. The completion of the centering movement is detected through position feedback, after which the locking mechanism secures the aircraft for internal transportation.

The attitude-adjustment platform returns to its reference position before the lifting mechanism begins the downward movement. During descent, the platform stops at a designated intermediate position to allow the propeller-positioning mechanism to operate. This operation is intended to prevent interference between the UAV propellers and the surrounding storage structure.

After confirmation that the propeller-positioning operation has been completed, the lifting platform continues toward the selected storage level. The roof is closed when the required clearance is available, and the horizontal transfer mechanism moves the parking platform toward the storage compartment.

The storage locking mechanism engages when the platform reaches its designated position. The connection between the parking platform and the horizontal transfer mechanism is then released. The transfer mechanism retracts, and the service connections are restored. The electrical connector is inserted into the corresponding UAV interface, followed by the replenishment connection.

The system subsequently checks whether a new deployment command has been received. If a valid command is present, the next operating sequence may be initiated. Otherwise, the mechanisms return to their standby positions and the recovery procedure is completed.

The recovery process therefore includes several operations that are not required during deployment, particularly touchdown confirmation, mechanical centering, locking, propeller positioning, and reconnection of the service interfaces. These additional operations must be considered when evaluating the total recovery cycle and the coordination requirements of the docking system. The principal deployment and recovery sequences are summarized in Figure 3. The flowchart identifies the main mechanical operations and the confirmation conditions required before the docking system proceeds to the next stage.

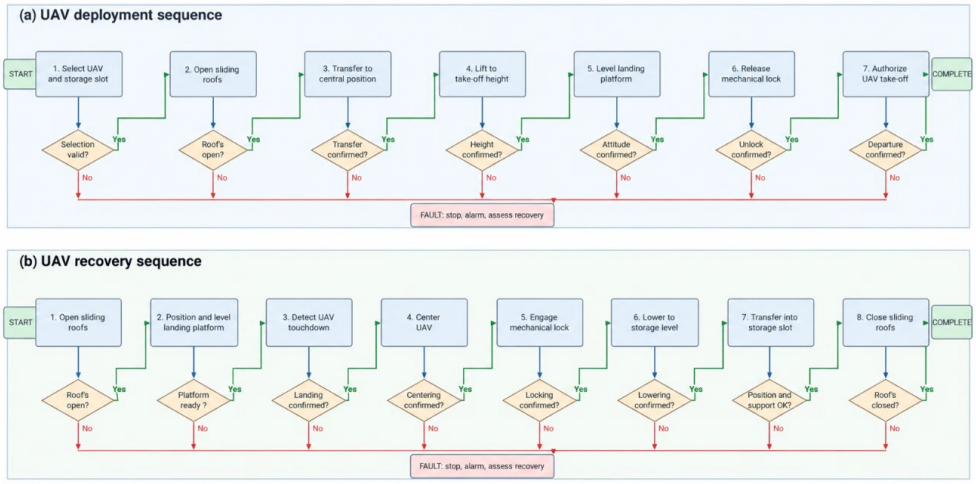


Figure 3. Operating sequences for unmanned aerial vehicle deployment and recovery

Note: UAV = unmanned aerial vehicle.

2.5 Operating Interlocks and Coordination Requirements

The deployment and recovery procedures demonstrate that reliable docking depends on the correct order of mechanical operations as well as the performance of individual actuators. Several critical interactions can be identified from the original design.

The first concerns the roof and lifting mechanism. The roof must provide sufficient clearance before the platform enters the external operating region. The second concerns the transfer and locking mechanisms. A parking platform must remain mechanically supported during the transfer of load between the storage compartment and the horizontal transfer assembly. The third concerns platform levelling and aircraft movement. Landing or take-off authorization should be associated with the required platform position and attitude condition. The fourth concerns centering and internal transportation. The aircraft must be secured before the parking platform is transferred into storage.

These requirements can be represented through a set of operating interlocks, as summarized in Table 1.

Table 1. Principal operating interlocks of the automated UAV docking system

Operation	Required Condition	Intended Response if Condition Is Not Satisfied
Parking platform extraction	Transfer mechanism engaged with parking platform	Inhibit storage-lock release
Vertical lifting through roof opening	Required roof clearance confirmed	Stop or inhibit lifting
Platform attitude adjustment	Landing platform at designated operating position	Inhibit levelling command
UAV take-off authorization	Platform levelling completed, roof clearance confirmed, and aircraft restraints released	Withhold launch permission
Mechanical centering after landing	Landing confirmation received	Inhibit centering movement
Transfer into storage	Aircraft secured and platform returned to reference attitude	Inhibit transfer
Reconnection of service interfaces	Parking platform locked in storage position	Inhibit connector actuation

Note: UAV = unmanned aerial vehicle.

The table identifies the principal logical dependencies inferred from the operating procedure. It is intended as a control-design representation rather than a record of completed functional safety testing.

For the subsequent analysis, the operation of the docking station can be divided into two interacting control tasks. The first is discrete coordination, which determines the permissible sequence of roof movement, lifting, transfer, centering, and locking. The second is continuous motion regulation, which determines the actuator movements required to adjust the landing platform orientation.

The discrete coordination task depends primarily on operating states, position feedback, and interlocking conditions. The continuous regulation task depends on platform geometry, attitude measurements, actuator displacement, and control response. Their interaction is important because the levelling mechanism must complete its required operation before the supervisory sequence can proceed to the aircraft launch or recovery stage.

The mechanical characteristics of the roof, landing platform, and centering assembly are examined in Section 3. The mathematical description of platform attitude adjustment and the coordination of the operating sequence are developed further in Section 4.

3 Mechanical Design and Mathematical Modelling

The mechanical design of the UAV docking station was developed around three principal subsystems: the protective sliding roof, the 3-DOF landing platform, and the mechanical centering and locking assembly. These subsystems serve different purposes during UAV deployment and recovery, but their operation is closely related. The roof provides the required clearance for external operations, the adjustable platform compensates for inclination of the supporting structure, and the centering assembly positions and secures the aircraft before internal transfer.

The present analysis retains the mechanical arrangements and engineering parameters of the original design. Mathematical models are introduced to describe transmission loading, platform motion, and operating constraints. The purpose is to establish a consistent mechanical basis for the coordinated control strategy discussed in Section 4 and the numerical evaluation presented in Section 5.

3.1 Sliding Roof Mechanism and Drive Modelling

3.1.1 Mechanical configuration

The protective roof consists of two horizontally translating panels arranged on opposite sides of the upper opening of the docking enclosure. Each panel moves outward along linear guides to provide clearance for the central lifting platform and UAV during deployment and recovery. The nominal travel distance of each panel is approximately 2000 mm.

Each roof panel is supported by two aluminium guide rails and four sliding blocks. The guide arrangement constrains the panel to linear translation and transfers its weight to the stationary enclosure frame. A brushless direct-current (DC) motor drives a rack-and-pinion transmission, with the motor mounted on the fixed structure and the rack attached to the moving panel. Rotation of the pinion is thereby converted into horizontal displacement.

The roof structure comprises a welded aluminium-alloy framework covered by a 2-mm aluminium sheet. A 40 mm aluminium silicate fibre insulation layer is installed within the framework to reduce heat exchange between the

enclosure and the surrounding environment. The supporting structure includes a central square-section beam and smaller transverse members. The external surface has a slightly raised central profile to facilitate drainage.

Weather protection is provided by a raised perimeter barrier and an overlapping joint between the roof panels. The mating edges incorporate complementary convex and concave profiles with a compressible sealing strip. This arrangement is intended to limit water ingress when the roof is closed. The original roof configuration is retained because its geometry and mass directly affect the transmission requirements and the operating clearance available to the lifting platform. The sliding roof assembly and its rack-and-pinion transmission are shown schematically in Figure 4. The arrangement illustrates the relationship between roof translation, guide support, and the transmission components used in the subsequent drive calculations.

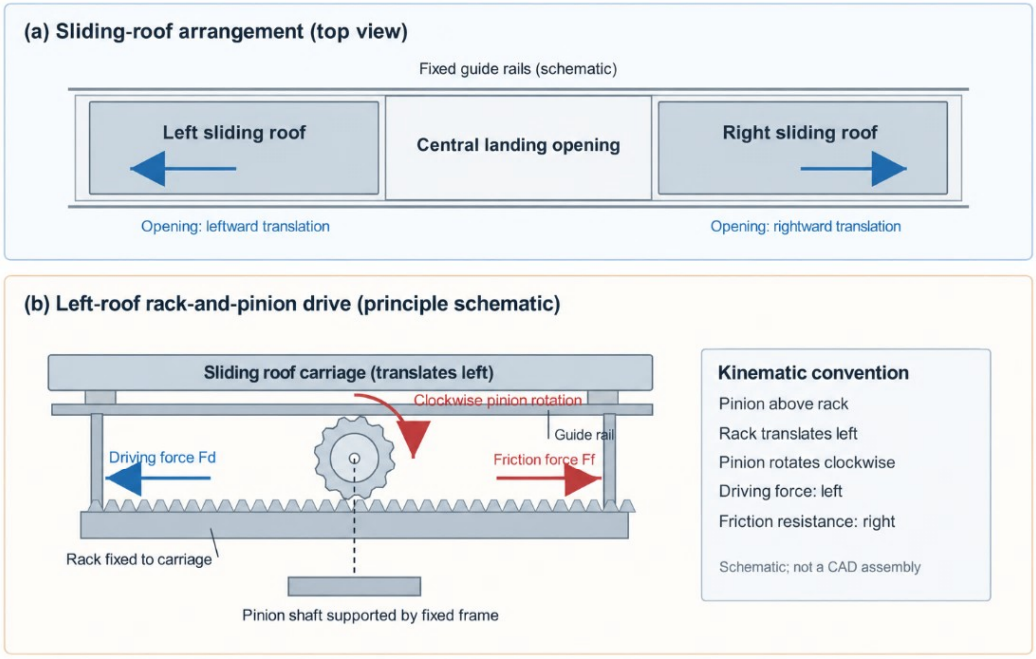


Figure 4. Mechanical arrangement and rack-and-pinion transmission of the sliding roof

Note: CAD = computer-aided design.

3.1.2 Friction load and transmission torque

The resistance to roof translation was estimated using a Coulomb friction model. Assuming horizontal motion and approximately constant friction, the nominal translational resistance is

$$F_r = \mu_r m_r g \tag{4}$$

where, F_r is the friction resistance, μ_r is the effective friction coefficient of the guide system, m_r is the mass of one moving roof panel, and g is gravitational acceleration.

The original design adopted a panel mass of 125 kg, an effective friction coefficient of 0.20, and a gravitational acceleration of 10 $\mathrm{m/s^2}$. The resulting resistance is

$$F_r = 0.20 \times 125 \times 10 = 250\ \mathrm{N} \tag{5}$$

The calculated resistance represents the nominal guide friction under horizontal, constant-speed conditions. Acceleration, wind loading, seal compression, and guide misalignment were excluded. The resulting drive torque therefore represents a reference operating requirement rather than the maximum load.

The roof uses a rack-and-pinion transmission with a module of 3 mm and 18 pinion teeth. The pinion pitch diameter is

$$d_p = m_p z_p = 3 \times 18 = 54\ \mathrm{mm} \tag{6}$$

where, m_p is the gear module and z_p is the number of teeth.

The torque required at the pinion is

$$T_p = F_r \frac{d_p}{2} \tag{7}$$

Substituting the design parameters gives

$$T_p = 250 \times 0.027 = 6.75\ \mathrm{N \cdot m} \tag{8}$$

The calculated value agrees with the original engineering estimate. It represents the nominal torque at the pinion rather than the motor-shaft torque. The actual motor torque depends on the gearbox ratio and mechanical transmission efficiency.

The theoretical displacement corresponding to one complete pinion revolution is

$$s_p = \pi d_p \tag{9}$$

For a pitch diameter of 54 mm, the displacement is approximately 169.65 mm per revolution.

3.1.3 Drive capacity and operating time

The original design selected a brushless DC motor with a reduction gearbox and electromagnetic brake. The reported gearbox ratio was 25:1, while the rated output torque T_{rated} was 24 $\mathrm{N \cdot m}$.

The nominal output torque margin is

$$K_T = \frac{T_{rated}}{T_p} = \frac{24}{6.75} \approx 3.56 \tag{10}$$

This ratio indicates the rated torque available relative to the calculated friction torque. It should not be treated as a verified safety factor because dynamic and environmental loads were not fully considered.

The roof velocity is related to the pinion rotational speed by

$$v_r = \frac{\pi d_p n_p}{60} \tag{11}$$

where, n_p is the pinion speed in revolutions per minute.

At an output speed of 60 r/min,

$$v_r = \frac{\pi \times 0.054 \times 60}{60} \approx 0.16965\ \mathrm{m/s} \tag{12}$$

For a nominal travel distance L_r of 2 m, the theoretical operating time is

$$t_r = \frac{L_r}{v_r} = \frac{2}{0.16965} \approx 11.79\ \mathrm{s} \tag{13}$$

The original design reported an operating time of approximately 12.5 s based on a nominal velocity of 0.16 m/s. The difference arises from the adopted speed assumptions. Both estimates describe idealized constant-speed motion and exclude acceleration, deceleration, and position-confirmation delays.

The mechanical output power at the pinion is

$$P_p = T_p \omega_p = T_p \frac{2\pi n_p}{60} \tag{14}$$

At the nominal pinion torque of 6.75 $\mathrm{N \cdot m}$ and output speed of 60 r/min, the required mechanical output power was approximately 42.4 W. The selected motor was reported to have a rated power of 150 W. This rating exceeds the calculated output requirement, although the adequacy of the motor and gearbox under transient and maximum-load conditions requires separate verification. The principal design parameters and calculated operating requirements of the sliding roof are summarized in Table 2.

The roof mechanism establishes an important clearance constraint in the docking sequence. Its position must be confirmed before the lifting platform enters the region requiring full roof opening. The nominal travel time also contributes to the overall deployment and recovery cycle, although the permissible overlap between roof and lifting movements depends on the mechanical clearance conditions.

3.2 Three-Degree-of-Freedom Landing Platform and Kinematic Modelling

3.2.1 Mechanical arrangement

The landing platform was designed to compensate for inclination of the docking enclosure and provide a controlled surface for UAV take-off and landing. The mechanism consists of an upper moving platform, a lower supporting platform, three servo-driven electric actuators, articulated joints, an attitude-sensing unit, and an electrical control assembly. The lower platform is attached to the central lifting structure, while the upper platform supports the UAV parking surface.

Table 2. Design parameters and calculated operating requirements of the sliding roof

Parameter	Value
Moving roof-panel mass	125 kg
Effective friction coefficient	0.20
Nominal friction resistance	250 N
Rack-and-pinion module	3 mm
Pinion teeth	18
Pinion pitch diameter	54 mm
Calculated pinion torque	6.75 N · m
Rated gearbox output torque	24 N · m
Nominal torque margin	3.56
Calculated mechanical output power	42.4 W
Reported motor power rating	150 W
Roof travel distance	2,000 mm
Reported opening time	Approximately 12.5 s

The upper and lower structural plates are manufactured from 10 mm aluminium plates. The electric actuators are connected to the moving platform through articulated joints, allowing their extension and retraction to change the platform position and orientation. Protective bellows are installed between the platform assemblies to reduce exposure of the internal components.

The electrical control system incorporates servo drives, attitude sensing, operating-state indication, overload protection, overcurrent protection, emergency shutdown, and actuator braking functions. The original design also specifies mechanical braking during abnormal operating conditions to prevent uncontrolled platform movement.

The platform is described as a 3-DOF mechanism. For the present analysis, its intended controlled motion is represented by vertical displacement, roll, and pitch. This representation is consistent with the levelling task, although the exact independent degrees of freedom depend on the actual joint configuration and must be confirmed from the detailed mechanical geometry. Figure 5 presents a schematic representation of the 3-DOF landing platform, including the upper moving platform, lower supporting structure, and three electric actuators. The controlled motions are represented by vertical displacement, roll, and pitch, consistent with the kinematic assumptions introduced in Section 3.2.2.

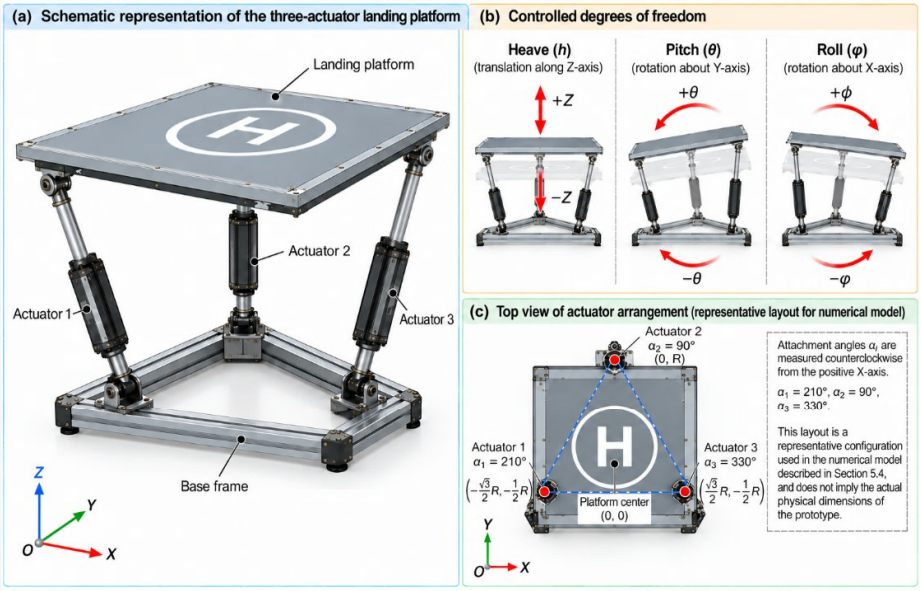


Figure 5. 3-degree-of-freedom landing platform and actuator arrangement

3.2.2 Coordinate systems and attitude representation

Two coordinate systems are introduced to describe platform motion. A fixed coordinate system $\{B\}$ is attached to the lower platform, while a moving coordinate system $\{P\}$ is attached to the upper landing surface.

The origin of $\{B\}$ is located at the reference centre of the lower platform. The origin of $\{P\}$ is defined at the corresponding reference point on the moving platform. The z-axis is taken as the nominal vertical direction, while the x- and y-axes describe the longitudinal and transverse directions of the landing surface.

The platform orientation is represented by the roll angle ϕ and pitch angle θ. Assuming that yaw remains constrained at its reference value, the rotation matrix is expressed as

$$R(\phi, \theta) = R_y(\theta) R_x(\phi) \tag{15}$$

where,

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

and

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \tag{16}$$

The translational position of the moving-platform reference point is represented by

$$p = \begin{bmatrix} 0 \\ 0 \\ h \end{bmatrix} \tag{17}$$

where, h denotes the vertical displacement relative to the fixed platform. The generalized coordinates are platform height, roll, and pitch, with yaw and independent horizontal translations assumed to be constrained. These assumptions must be checked against the actual joint configuration, particularly if the mechanism permits coupled horizontal motion.

3.2.3 Inverse kinematic model

The orientation of the landing surface is controlled through the extension and retraction of three electric actuators. The required actuator lengths depend on the attachment-point geometry and the desired platform pose.

Let a_i denote the position vector of the i-th actuator attachment point on the fixed platform, expressed in $\{B\}$, and let b_i denote the corresponding attachment point on the moving platform, expressed in $\{P\}$.

For a compatible three-actuator parallel arrangement, the actuator vector is

$$l_i = p + R(\phi, \theta) b_i - a_i, \quad i = 1, 2, 3 \tag{18}$$

The actuator length is then obtained as

$$L_i = \| p + R(\phi, \theta) b_i - a_i \| \tag{19}$$

The required actuator displacement relative to a reference configuration is

$$\Delta L_i = L_i - L_{i0} \tag{20}$$

where, L_{i0} is the actuator length at the reference position.

Eqs. (18)–(20) establish the inverse kinematic relationship between platform pose and actuator displacement. They provide the mathematical basis for converting the desired platform attitude into actuator position commands.

The inverse kinematic relationships describe the platform under the assumed joint and motion constraints. The attachment-point coordinates are retained as geometric parameters because the complete mechanism dimensions are unavailable. The model must be checked against the actual joint arrangement before it is used to predict actuator displacement or platform motion.

3.2.4 Attitude feedback and mechanical limits

The platform attitude is obtained from the installed inertial sensing arrangement. The original design describes the use of gyroscope feedback, inverse kinematic calculation, and servo-driven actuator adjustment to maintain the required landing-platform orientation.

The attitude error is represented by

$$e_a(t) = \begin{bmatrix} \phi_d(t) - \phi_m(t) \\ \theta_d(t) - \theta_m(t) \end{bmatrix} \tag{21}$$

where, ϕ_d and θ_d are the desired roll and pitch angles, and ϕ_m and θ_m are the measured angles.

For stationary levelling, the desired orientation corresponds to a horizontal landing surface. In a moving installation, the reference attitude may depend on the disturbance conditions and the adopted control strategy.

The platform control arrangement includes high-pass and low-pass filtering channels associated with washout-based motion processing. These filters provide a means of separating motion components according to frequency. Their effect on platform levelling depends on the filter settings, actuator response, and feedback controller. Since these parameters have not been established for the present model, no improvement in dynamic response is assumed.

The actuator motion is subject to stroke and velocity constraints:

$$L_{i,\min} \le L_i(t) \le L_{i,\max}, \quad \left|\dot{L}_i(t)\right| \le v_{i,\max} \tag{22}$$

where, $L_{i,\min}$, $L_{i,\max}$, and $v_{i,\max}$ denote the minimum actuator length, maximum actuator length, and allowable actuator velocity, respectively.

These constraints define the mechanically permissible operating region of the landing platform. Their numerical limits must be obtained from the selected actuator specifications and platform geometry.

3.3 Mechanical Centering and Locking Mechanism

3.3.1 Mechanical configuration

The centering and locking mechanism was designed to correct residual positioning errors after UAV landing and to secure the aircraft during internal transfer and storage. The assembly is integrated into the parking platform and consists of transverse and longitudinal centering bars, trapezoidal lead screws, linear guide rails, a central transmission shaft, mechanical connectors, and locking elements. The centering bars are arranged at different heights to prevent mechanical interference during their movement.

The longitudinal centering bars primarily contact the UAV landing gear and guide the aircraft toward the designated central position. The transverse bars incorporate four mechanical locking claws that engage with the reinforced members of the landing gear after the centering operation. This arrangement combines positional correction and mechanical restraint within a common assembly, reducing the need for separate positioning and securing mechanisms.

The original design uses a single drive input connected to a central transmission shaft. The transmitted motion drives the lead-screw assemblies and moves the centering bars toward the platform centre. Linear guides constrain the bars to their intended paths, while position switches and encoder feedback are used to confirm the completion of movement. Once the aircraft reaches the required position, the locking elements engage to prevent excessive displacement during subsequent lifting and horizontal transfer.

The centering mechanism must accommodate the positional deviation remaining after touchdown. Its operating range is therefore determined by the allowable landing-position error, the available stroke of the centering bars, and the geometry of the UAV landing gear. The mechanical locking function must also remain effective when the docking station is subjected to transportation-induced vibration or inclination. These requirements link the centering mechanism directly to the supervisory interlocks described in Section 2. The principal components of the centering and locking mechanism are illustrated in Figure 6. The arrangement shows the transverse and longitudinal centering bars, lead-screw transmission, linear guides, and locking elements used to position and secure the UAV after landing.

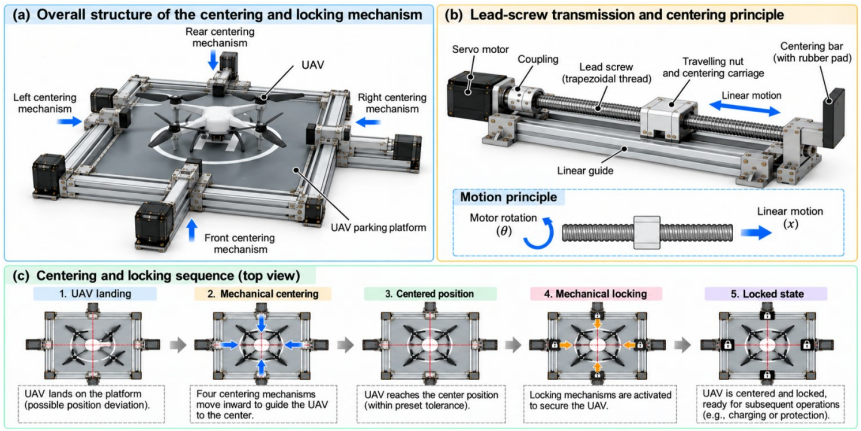


Figure 6. Mechanical centering and locking mechanism of the UAV docking system

Note: UAV = unmanned aerial vehicle.

3.3.2 Centering resistance and design axial load

The force required to move the UAV across the parking surface was estimated from the frictional resistance between the aircraft landing gear and the supporting platform. Under the assumption of horizontal movement and approximately constant friction, the nominal resistance is

$$F_c = \mu_c m_c g \tag{23}$$

where, F_c is the centering resistance, μ_c is the effective friction coefficient between the landing gear and parking surface, m_c is the UAV mass, and g is gravitational acceleration.

The original engineering design adopted a UAV mass of 130 kg, a friction coefficient of 0.25, and a gravitational acceleration of 10 $\mathrm{m/s^2}$. The resulting resistance was

$$F_c = 0.25 \times 130 \times 10 = 325 \text{ N} \tag{24}$$

This value represents the estimated resistance under the specified assumptions. Actual centering loads may vary with landing-gear material, contact conditions, surface contamination, aircraft inclination, and the distribution of contact forces between the centering bars.

To account for uneven loading and operational uncertainty, the original design adopted a load factor of 3. The corresponding design axial load was calculated as

$$F_s = k_s F_c = 3 \times 325 = 975 \text{ N} \tag{25}$$

where, k_s is the adopted load factor.

The design axial load of 975 N was assigned to each loaded screw for preliminary torque estimation. The actual force distribution depends on the contact arrangement and the number of screws carrying the load simultaneously.

The centering mechanism was specified to use a trapezoidal lead screw with a nominal diameter of 10 mm and a reported lead of 4 mm. A thread pitch of 2 mm would correspond to a two-start configuration. The complete thread designation and pitch diameter must be confirmed against the selected screw specification before the transmission efficiency and self-locking condition can be evaluated.

3.3.3 Lead-screw drive torque and power requirements

The torque required to drive a lead screw depends on the axial load, screw lead, and mechanical efficiency. Using the simplified power-transmission relationship, the required torque for one loaded screw is expressed as

$$T_s = \frac{F_s p_s}{2\pi \eta_s} \tag{26}$$

where, T_s is the lead-screw driving torque, F_s is the axial design load, p_s is the screw lead, and η_s is the assumed transmission efficiency.

For F_s = 975 N, p_s = 0.004 m, and the originally adopted efficiency of 0.55, the calculated torque is

$$T_s = \frac{975 \times 0.004}{2\pi \times 0.55} \approx 1.13 \text{ N} \cdot \text{m} \tag{27}$$

The original design considered a common drive motor transmitting motion to two lead-screw assemblies. For two equally loaded lead screws, the motor torque can be estimated by accounting for the motor-to-screw speed ratio and the additional transmission efficiency:

$$T_m = \frac{2T_s}{i_t \eta_t} \tag{28}$$

For two equally loaded lead screws, the required motor torque is given by Eq. (28), where i_t = n_m / n_s is the motor-to-screw speed ratio and η_t is the additional transmission efficiency. The estimated torque of 4.51 $\mathrm{N \cdot m}$ corresponds to the assumed condition i_t = 1. The actual motor torque must be recalculated if the transmission ratio or load distribution differs from the assumed values.

The original design specified a motor operating speed of 1,200 r/min. The corresponding mechanical power at the motor shaft is

$$P_m = T_m \frac{2\pi n_m}{60} \tag{29}$$

where, n_m is the motor rotational speed.

Under the assumed unit motor-to-screw speed ratio, the estimated motor torque of 4.51 $\mathrm{N \cdot m}$ at 1,200 r/min corresponds to a mechanical shaft power of approximately 0.57 kW. The reported centering time of 10 s remains a design estimate because the actual transmission ratio and centering stroke have not been confirmed.

Table 3 summarizes the design parameters and estimated drive requirements of the centering mechanism under the stated transmission assumptions.

Table 3. Design parameters and calculated requirements of the centering mechanism

Parameter	Value
UAV mass	130 kg
Effective friction coefficient	0.25
Nominal centering resistance	325 N
Adopted load factor	3
Design axial load	975 N
Reported lead-screw specification	Nominal diameter 10 mm; lead 4 mm; pitch 2 mm (unverified)
Reported screw lead	4 mm
Reported thread pitch	2 mm
Assumed lead-screw efficiency for torque estimation	0.55
Assumed additional transmission efficiency	0.50
Calculated torque per loaded screw	$1.13\ \mathrm{N \cdot m}$
Calculated common-drive motor torque	$4.51\ \mathrm{N \cdot m}$
Reported motor operating speed	1,200 r/min
Calculated motor-shaft mechanical power	Approximately 0.57 kW
Reported centering time	Approximately 10 s

Note: UAV = unmanned aerial vehicle.

3.3.4 Positioning and mechanical locking constraints

The centering mechanism performs two related functions: correcting the UAV position and restraining the aircraft after positioning. These functions impose different mechanical requirements.

During centering, the bars must provide sufficient force to overcome the resistance between the aircraft landing gear and the parking surface. The bars must also move within their available stroke without interfering with the landing gear or other structural components. The permissible movement of an individual centering bar can be expressed as

$$x_{c,\min} \leq x_c(t) \leq x_{c,\max} \tag{30}$$

where, $x_c(t)$ is the bar displacement and $x_{c,\min}$ and $x_{c,\max}$ are its mechanical travel limits.

After positioning, the locking elements must restrain the aircraft against unwanted movement. The original design selected a trapezoidal lead-screw transmission partly because of its potential resistance to back-driving. However, self-locking cannot be assumed solely from the screw type.

For a conventional power screw, the possibility of self-locking depends on the relationship between the screw lead angle and the effective friction angle. A simplified condition is

$$\tan \lambda < \mu_{eq} \tag{31}$$

where, λ is the screw lead angle and μ_{eq} is the effective friction coefficient accounting for the screw-thread contact geometry.

The self-locking condition depends on the screw lead angle and effective thread friction. The assumed lead-screw efficiency of 0.55 was used only for preliminary torque estimation and does not demonstrate self-locking. Since a relatively high forward efficiency may be incompatible with self-locking under the same friction conditions, the mechanical locking claws are treated as the primary positive restraint. Their engagement must be confirmed before internal transfer.

3.4 Mechanical Design Verification and Operating Constraints

3.4.1 Consistency of the mechanical design calculations

The mechanical design calculations were reviewed to establish a consistent basis for subsequent coordinated motion analysis. The original manuscript reported the principal operating parameters of the roof and centering mechanisms, including their friction loads, driving torques, motor selections, and estimated operating times. Most of the nominal force and torque values can be reproduced from the stated design assumptions.

For the sliding roof, the adopted mass of 125 kg and friction coefficient of 0.20 produced a calculated resistance of 250 N. The rack-and-pinion geometry then gave a nominal driving torque of 6.75 $\mathrm{N} \cdot \mathrm{m}$. The reported gearbox output torque of 24 $\mathrm{N} \cdot \mathrm{m}$ provided a nominal torque margin of approximately 3.56 relative to this calculated requirement.

The roof operating time requires a distinction between the reported nominal speed and the speed calculated from the pinion geometry. At 60 r/min, the theoretical linear speed was approximately 0.16965 m/s, corresponding to a 2 m travel time of 11.79 s. The original operating time of 12.5 s corresponded to a nominal speed of 0.16 m/s. Both values are reasonable within their respective assumptions, but the actual operating time depends on acceleration, deceleration, and the position-control sequence.

For the centering mechanism, the adopted UAV mass of 130 kg and friction coefficient of 0.25 gave a nominal resistance of 325 N. Applying the original load factor of 3 produced a design axial load of 975 N. The simplified lead-screw transmission model yielded a torque of approximately 1.13 $\mathrm{N} \cdot \mathrm{m}$ per loaded screw. Assuming two equally loaded screws, a unit motor-to-screw speed ratio, and an additional transmission efficiency of 0.50, the estimated motor torque was approximately 4.51 $\mathrm{N} \cdot \mathrm{m}$. At the assumed motor speed of 1,200 r/min, the corresponding shaft power was approximately 0.57 kW. Both estimates require revision if the actual transmission ratio differs from unity.

These calculations establish nominal design requirements but do not constitute experimental verification. In particular, the assumed friction coefficients, load-sharing conditions, and transmission efficiencies must be assessed against the actual mechanism configuration before the calculated margins can be used to establish operational reliability.

3.4.2 Mechanical operating constraints

The three principal mechanisms impose different constraints on the docking sequence.

The roof mechanism is primarily constrained by its available travel, driving torque, and opening clearance. The lifting platform must not enter the roof opening until the required clearance has been established. Although roof opening and platform lifting may overlap during part of the operating cycle, their relative motion must remain within the collision-free region defined by the enclosure geometry.

The landing platform is constrained by actuator stroke, allowable actuator velocity, mechanical joint limits, and the reachable orientation range. The inverse kinematic model developed in Section 3.2 provides a means of relating the desired platform pose to actuator lengths. However, the attainable operating region depends on the actual attachment-point coordinates and actuator specifications. Platform levelling must therefore be completed within the mechanically permissible workspace before the system authorizes the relevant UAV operation.

The centering mechanism is constrained by the available bar stroke, the geometry of the aircraft landing gear, the transmission force capacity, and the locking arrangement. The centering bars must not be activated before the aircraft has landed and the required operating conditions have been confirmed. Similarly, the parking platform must not be transferred into storage before the centering and locking operations have been completed.

The principal mechanical constraints and their implications for coordinated operation are summarized in Table 4.

Table 4. Mechanical constraints and implications for coordinated docking operations

Subsystem	Principal Constraint	Implication for Coordinated Operation
Sliding roof	Travel limit and required opening clearance	Lifting movement must remain within the available clearance
Roof drive	Available torque and operating speed	Opening and closing commands must respect drive capability
3-DOF landing platform	Actuator stroke and velocity limits	Attitude commands must remain within the reachable workspace
Platform joints	Mechanical articulation limits	Excessive platform inclination must be prevented
Centering assembly	Bar stroke and available driving force	Position correction must remain within the permissible range
Locking mechanism	Engagement and holding capability	Internal transfer requires locking confirmation
Shared transfer system	Position and mechanical engagement conditions	Load transfer must follow the required locking sequence

Note: DOF = degree of freedom.

The constraints listed in Table 4 provide the mechanical basis for the supervisory interlocks introduced in Section 2. They also define the conditions that must be respected by the coordinated motion control strategy developed in Section 4.

3.4.3 Parameter uncertainty and design margins

The nominal load calculations depend on several parameters that may vary during operation. For the roof, the effective friction coefficient can change with guide condition, temperature, contamination, and mechanical alignment. The roof may also experience additional resistance from sealing contact or external wind loading. For the centering mechanism, the contact resistance depends on the landing-gear material, surface condition, aircraft mass, and the distribution of forces among the centering bars.

These uncertainties affect the interpretation of the calculated drive margins. A motor selected solely from the nominal friction load may provide adequate capacity under the assumed conditions but insufficient capacity when additional resistance is present. The design evaluation should therefore consider a range of plausible operating parameters rather than relying exclusively on a single nominal value.

The relationship between load variation and driving torque can be examined using the analytical models developed in Sections 3.1 and 3.3. For the roof, the required pinion torque varies directly with the effective friction coefficient and panel mass when the gear geometry remains unchanged. For the centering mechanism, the lead-screw torque varies with the axial load and inversely with the assumed transmission efficiency.

These relationships provide the basis for the parameter sensitivity analyses presented in Section 5. The purpose of those analyses is to identify the influence of uncertain mechanical parameters on the required drive capacity, rather than to claim that the complete docking system has been experimentally validated.

3.4.4 Implications for coordinated motion control

The mechanical analysis establishes three requirements for coordinated docking operation.

First, the supervisory controller must account for the operating positions of the roof, lifting mechanism, and transfer assembly. These mechanisms occupy a shared working space, and their movements must be authorized according to the available clearance and mechanical engagement conditions.

Second, the landing-platform controller must generate actuator commands that satisfy the platform kinematic relationships and physical operating limits. The desired platform attitude cannot be considered independently of actuator stroke, joint articulation, and the mechanical workspace.

Third, the centering and locking mechanism must complete its required positioning and restraint operations before the aircraft is transferred into storage. The completion of motor movement alone does not necessarily establish that the aircraft has been securely locked; the corresponding confirmation signals must also be considered.

These requirements indicate that the mechanical and control subsystems cannot be evaluated entirely in isolation. The roof and centering calculations establish actuator loading and timing requirements, while the platform kinematic model defines the relationship between desired attitude and actuator motion. Together, they provide the mechanical constraints for the coordinated control formulation developed in Section 4.

The subsequent numerical evaluation in Section 5 uses the available engineering parameters to examine the sensitivity of the roof and centering mechanisms to changes in load and transmission conditions. Quantitative evaluation of the 3-DOF platform response requires additional geometric and dynamic parameters, which remain necessary for establishing the control performance of the physical docking system.

4 Coordinated Motion Control

The mechanical analysis in Section 3 established the drive requirements and operating limits of the principal docking mechanisms. These constraints must be considered together during unattended UAV deployment and recovery. Although the sliding roof, lifting assembly, landing platform, and centering mechanism are driven by separate actuators, their movements share a common operating space and must follow a prescribed sequence. A suitable control arrangement must therefore address both continuous platform motion and the discrete coordination of mechanical operations.

The original docking-station design incorporated attitude sensing, inverse kinematic calculations, servo-driven platform adjustment, position feedback, and sequential interlocks. Building on these functions, this section formulates a coordinated control framework that separates platform attitude regulation from supervisory sequence management. The formulation is intended to support subsequent numerical analysis and controller implementation. It does not imply that all the control laws introduced below have already been implemented or experimentally validated on the physical docking station.

4.1 Control Objectives and Supervisory Structure

The docking control system has two principal responsibilities. The first is to regulate the orientation of the 3-DOF landing platform so that the UAV can take off from or land on a surface with the required attitude. The second is to coordinate the movements of the roof, lifting mechanism, horizontal transfer assembly, centering bars, and mechanical locks without violating their operating constraints.

These responsibilities are handled at different levels of the proposed control architecture. The supervisory controller manages the operating sequence, evaluates sensor and interlock signals, and authorizes transitions between mechanical states. The platform controller processes attitude measurements and generates actuator displacement commands. Individual motor drives execute the required movements and report their operating conditions to the supervisory level.

For analytical purposes, the control architecture is divided into three functional layers.

The supervisory layer receives launch and recovery requests, selects the relevant storage compartment, and determines the permitted operating sequence. It also exchanges status and authorization signals with the UAV. The motion-control layer generates commands for the roof, lifting assembly, horizontal transfer mechanism, landing platform, and centering system. The sensing and protection layer provides position feedback, actuator status, attitude measurements, and abnormal-condition signals.

The principal control objectives are to maintain the required landing-platform attitude, avoid interference between moving mechanisms, preserve mechanical support during platform transfer, and prevent internal transportation before the aircraft has been secured.

The control framework also needs to account for actuator limitations. The platform controller must not command actuator lengths outside the available stroke, while the supervisory controller must prevent operations that conflict with the mechanical conditions identified in Table 4.

The overall control arrangement is therefore based on a combination of continuous attitude regulation and discrete-event coordination. The former determines the actuator movements required for platform levelling; the latter determines when each mechanical operation may begin or end. The interaction between continuous platform attitude regulation and discrete supervisory coordination is illustrated in Figure 7. The framework shows how attitude feedback, actuator commands, operating-state information, and safety interlocks are combined to govern the docking sequence.

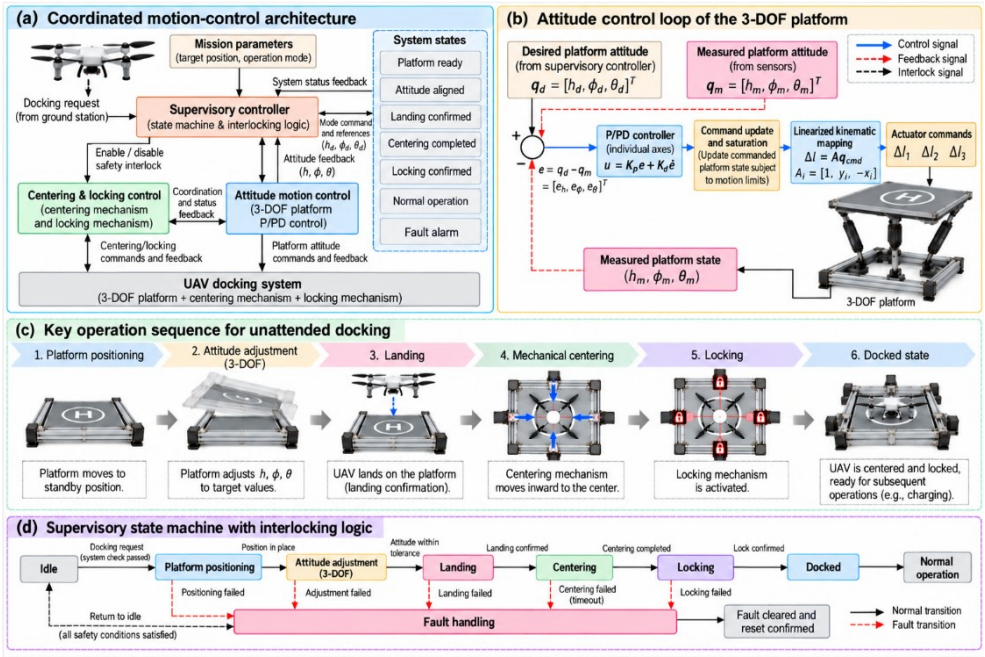


Figure 7. Coordinated motion-control and supervisory interlocking framework for unattended unmanned aerial vehicle docking

Note: UAV = unmanned aerial vehicle; DOF = degree of freedom; P = proportional; PD = proportional-derivative.

4.2 Attitude Measurement and Reference Generation

4.2.1 Platform attitude measurement

The original platform design uses an inertial sensing arrangement to obtain attitude-related feedback during levelling. Gyroscope measurements provide angular-rate information, which can be used together with an appropriate attitude-estimation method to determine the orientation of the landing surface.

For the control formulation, the measured platform attitude is represented by

$$\eta_{\mathrm{m}}(t) = \begin{bmatrix} \phi_{\mathrm{m}}(t) \\ \theta_{\mathrm{m}}(t) \end{bmatrix} \tag{32}$$

where, ϕ_{m} and θ_{m} are the measured roll and pitch angles.

The desired attitude is represented by

$$\eta_{\mathrm{d}}(t) = \begin{bmatrix} \phi_{\mathrm{d}}(t) \\ \theta_{\mathrm{d}}(t) \end{bmatrix} \tag{33}$$

The attitude error is obtained from the difference between the desired and measured orientations:

$$e_{\eta}(t) = \eta_{\mathrm{d}}(t) - \eta_{\mathrm{m}}(t) \tag{34}$$

This error provides the feedback quantity used to determine the required platform correction.

Gyroscope measurements provide angular velocity rather than an absolute attitude reference. Platform roll and pitch therefore require an appropriate attitude-estimation method, such as the combination of gyroscope and accelerometer measurements. The subsequent control formulation assumes that estimates of both angles are available, without prescribing a particular sensor-fusion algorithm.

4.2.2 Reference attitude and disturbance compensation

For a stationary docking station installed on an inclined support, the principal levelling objective is to maintain the upper landing surface approximately horizontal with respect to gravity. The corresponding reference roll and pitch angles are zero in the adopted world-referenced convention.

For a docking station installed on a moving vehicle or vessel, the control requirement is more complicated. The supporting structure may undergo time-varying roll and pitch disturbances, while the platform actuators must compensate for these movements within their available stroke and speed limits.

The desired platform attitude must therefore be generated with reference to the intended operating condition. In a stationary levelling task, the controller can use a constant horizontal reference. Under dynamic excitation, the reference-generation process may also need to account for the frequency content of the supporting-platform motion.

The original engineering design introduced a washout-based control arrangement incorporating high-pass and low-pass filtering channels. Such filters can separate motion components according to their frequency characteristics. In the present study, this arrangement is retained as part of the proposed reference-generation framework.

However, the washout algorithm should not be interpreted as a complete disturbance-rejection controller by itself. Its effect depends on the selected filter characteristics, the actuator bandwidth, and the relationship between the supporting-platform motion and the desired landing-surface orientation.

For subsequent numerical analysis, the reference-generation function can be represented generally as

$$\eta_{\mathrm{d}}(t) = \mathcal{W}\left[\eta_{\mathrm{ref}}(t), d(t)\right] \tag{35}$$

where, $\mathcal{W}$ denotes the reference-generation or washout-processing function, η_{ref} is the nominal reference attitude, and $d(t)$ represents the measured or estimated disturbance information.

Eq. (35) describes the intended reference-generation arrangement at the functional level. Since the washout filter structure and coefficients were not specified, this arrangement was not implemented in the numerical simulations presented in Section 5.4. Those simulations focused on P and PD attitude feedback without washout-based reference processing.

4.3 Platform Levelling and Actuator Coordination

4.3.1 Attitude correction strategy

The 3-DOF landing platform adjusts its orientation through coordinated extension and retraction of three servo-driven electric actuators. The inverse kinematic relationships developed in Section 3.2 convert a desired platform pose into corresponding actuator lengths.

A candidate feedback-control strategy is introduced to relate the measured attitude error to the commanded platform correction. For a PD formulation, the attitude correction command may be expressed as

$$u_{\eta}(t) = K_{\mathrm{p}} e_{\eta}(t) + K_{\mathrm{d}} \dot{e}_{\eta}(t) \tag{36}$$

where, K_{p} and K_{d} are P and derivative gain matrices, respectively.

Eq. (36) describes a candidate feedback-control law rather than the verified controller of the manufactured docking system. The gain values adopted for the representative numerical study are specified in Section 5.4.

In Eq. (36), the feedback output is interpreted as an attitude correction rather than a direct actuator command. The P and derivative gains must be defined consistently with the units of attitude and angular velocity. The correction is combined with the nominal reference to obtain a feasible platform attitude command, which is then converted into actuator lengths through inverse kinematics. The sampling interval, gain values, and command-limiting procedure must be specified before implementation.

The corresponding actuator command vector is represented by

$$L_{\mathrm{d}}(t) = \begin{bmatrix} L_{1,\mathrm{d}}(t) \\ L_{2,\mathrm{d}}(t) \\ L_{3,\mathrm{d}}(t) \end{bmatrix} = f_{\mathrm{IK}}\left(h_{\mathrm{d}}(t), \phi_{\mathrm{d}}^{*}(t), \theta_{\mathrm{d}}^{*}(t)\right) \tag{37}$$

where, f_{IK} is the inverse kinematic mapping developed in Section 3.2, h_{d} is the desired platform height, and ϕ_{d}^{*} and θ_{d}^{*} are the corrected attitude commands.

The commanded lengths must remain within the actuator stroke limits specified in Eq. (22). The controller must also account for actuator speed limits to prevent commands that cannot be executed within the required operating time.

4.3.2 Actuator synchronization and motion constraints

The three electric actuators must operate in a coordinated manner because their individual displacements collectively determine the landing-platform pose. A command applied to one actuator changes the geometric relationship between the moving and fixed platforms and may affect the load distribution among the remaining actuators.

The actuator position error is represented by

$$e_L(t) = L_{\mathrm{d}}(t) - L_{\mathrm{m}}(t) \tag{38}$$

where, $L_{\mathrm{m}}(t)$ contains the measured actuator lengths.

A suitable servo-control arrangement seeks to reduce the actuator position errors while respecting the mechanical constraints. In practice, each actuator may have an individual position-control loop, while the platform-level controller coordinates the desired actuator trajectories through the inverse kinematic model.

The coordination requirement is not that all actuators move through identical distances. Different platform attitudes generally require different actuator displacements. Instead, the actuator movements must remain consistent with the desired platform geometry throughout the motion.

The commanded actuator trajectories should also be checked against the permissible workspace. If the desired platform pose requires an actuator to exceed its stroke or velocity limit, the command must be modified or rejected. This condition is particularly important when the docking station is subjected to large inclination disturbances.

The platform should not be declared ready for UAV operation solely because the actuator motors have stopped. The supervisory controller must also confirm that the platform has reached the required operating position and that the measured attitude error is within the specified tolerance.

A representative levelling-completion condition can be written as

$$E_{\mathrm{level}} = \left(|\phi_{\mathrm{m}} - \phi_{\mathrm{d}}| \leq \varepsilon_{\phi}\right) \wedge \left(|\theta_{\mathrm{m}} - \theta_{\mathrm{d}}| \leq \varepsilon_{\theta}\right) \tag{39}$$

where, ε_{ϕ} and ε_{θ} are the allowable roll and pitch errors.

The numerical values of these tolerances must be determined from the landing requirements and the actual platform capability. They were not provided in the original design.

4.3.3 Interaction between levelling and the docking sequence

Platform levelling forms one stage of the overall docking operation. Its completion determines whether the system can proceed to UAV take-off or landing authorization.

During deployment, the platform is first transferred to the external operating position. The attitude-adjustment mechanism then corrects the landing-surface orientation. After the required position and attitude conditions have been confirmed, the supervisory controller may proceed with the remaining launch-preparation operations.

During recovery, the platform is similarly positioned and levelled before landing permission is issued. Following touchdown and aircraft centering, the platform returns to its reference configuration before the aircraft is transferred into storage.

This interaction means that platform control cannot be evaluated solely through attitude tracking. Its operating time, completion conditions, and failure behaviour also influence the total docking sequence. A platform controller with acceptable steady-state accuracy but excessive settling time may delay deployment or recovery, while a controller that produces actuator saturation may prevent the required operating state from being reached.

These considerations provide the basis for the control-performance evaluation criteria introduced in Section 4.6.

4.4 Discrete-Event Coordination of Docking Mechanisms

4.4.1 Supervisory state management

The docking station operates through a sequence of mechanical actions governed by position feedback and interlocking conditions. The supervisory controller manages the operating states of the roof, lifting platform, horizontal transfer assembly, centering mechanism, and mechanical locks. Each mechanism can enter its next operating state only after the required conditions have been confirmed.

For a given operation, the authorization condition is expressed as

$$G_j = \bigwedge_{i=1}^{n_j} c_{j,i} \tag{40}$$

where, G_j denotes the authorization condition for operation j, $c_{j,i}$ represents an individual interlock condition, and n_j is the number of mandatory conditions. This formulation describes the supervisory logic without prescribing a particular controller implementation.

4.4.2 Coordination during unmanned aerial vehicle deployment

During deployment, the selected UAV is disconnected from its service interfaces, and the parking platform is transferred from storage to the central lifting assembly. The transfer mechanism must engage with the parking platform before the storage lock is released.

The lifting platform then moves toward the external operating position while the roof opens. These movements may overlap provided that sufficient mechanical clearance is maintained:

$$C_{\mathrm{rl}}\left(x_{\mathrm{r}}, z_{\mathrm{l}}\right) \geq C_{\min} \tag{41}$$

where, C_{rl} is the available clearance determined by roof displacement x_{r} and lifting-platform position z_{l}, and $C_{\min}$ is the required minimum clearance.

Once the platform reaches its operating height, the levelling controller adjusts its attitude. Take-off is authorized only after the required platform position, attitude, roof clearance, and mechanical release conditions have been confirmed. Following aircraft departure, the platform returns to its reference position and the mechanisms complete the storage and reset sequence.

4.4.3 Coordination during unmanned aerial vehicle recovery

During recovery, an available parking platform is transferred to the central lifting area and raised to the landing position. Roof opening and platform levelling must be completed before landing authorization is issued.

Following touchdown, the centering bars correct the aircraft position, and the mechanical locking elements secure the landing gear. Completion of centering and confirmation of locking are treated as separate requirements. Internal transfer is authorized only when

$$G_{\mathrm{transfer}} = E_{\mathrm{position}} \wedge E_{\mathrm{lock}} \wedge E_{\mathrm{platform}} \tag{42}$$

where, the three conditions represent confirmed aircraft positioning, mechanical locking, and the required platform transfer configuration.

The platform subsequently returns to its reference attitude and descends. After the intermediate propeller-positioning operation, the aircraft is transferred into storage, secured, and reconnected to its service interfaces.

4.4.4 Coordination between continuous control and discrete operations

The docking system combines continuous platform attitude regulation with discrete mechanical operations. The supervisory controller links these functions by monitoring actuator positions, platform attitude, and mechanical interlocks.

Completion of an actuator movement does not necessarily indicate that the required platform attitude has been achieved. Accordingly, operating authorization depends on the combined confirmation of positioning, levelling, and interlocking conditions:

$$E_{\mathrm{op}} = E_{\mathrm{position}} \wedge E_{\mathrm{level}} \wedge E_{\mathrm{interlock}} \tag{43}$$

This arrangement allows compatible operations, such as roof opening and early-stage lifting, to overlap while preventing movements that violate mechanical clearance or locking requirements. The coordination is therefore governed by operating conditions rather than fixed time delays alone.

4.5 Fault Handling and Safe-State Behaviour

4.5.1 Fault detection and operating interruption

Unattended docking requires abnormal conditions to be identified before they cause mechanical interference or loss of aircraft restraint. The original design incorporates position monitoring, movement timeouts, electrical protection, emergency shutdown, and actuator braking.

Potential faults include incomplete roof opening, transfer-position errors, unsuccessful locking, excessive platform inclination, actuator overload, and communication failure. A timeout condition for operation j is defined as

$$E_{\mathrm{timeout},j} = \begin{cases} 1, & t_j > t_{j,\max} \text{ and completion is unconfirmed} \\ 0, & \text{otherwise} \end{cases} \tag{44}$$

where, t_j is the elapsed time since operation j was initiated, and $t_{j,\max}$ is its prescribed completion limit. A timeout is registered only when the operation remains incomplete after this limit has been exceeded.

4.5.2 Safe-state management

The response to a fault depends on the mechanical configuration at the time of interruption. Removing actuator power alone may not provide a safe condition when the lifting or transfer mechanism is supporting an aircraft.

The supervisory controller should inhibit incompatible movements, preserve existing mechanical support where possible, and prevent further UAV operations until the fault has been resolved.

The principal fault conditions and the corresponding supervisory responses are summarized in Table 5.

Table 5. Representative fault conditions and proposed supervisory responses

Fault Condition	Detection Basis	Proposed Response
Incomplete roof opening	Position feedback or timeout	Inhibit lifting into the roof-clearance region
Transfer-position error	Position switch or encoder	Prevent locking release or further transfer
Levelling failure	Attitude error or actuator limit	Withhold UAV operating authorization
Incomplete centering	Position feedback or timeout	Prevent transfer into storage
Locking failure	Lock-position feedback	Maintain support and inhibit transfer
Actuator overload	Drive protection signal	Stop the affected movement and retain mechanical restraint
Communication failure	Missing status confirmation	Suspend operations requiring UAV authorization
Emergency shutdown	Emergency-stop input	Inhibit commands and initiate the equipment-specific stopping procedure

Note: UAV = unmanned aerial vehicle.

These responses represent proposed supervisory requirements and require verification on the physical system.

4.5.3 Recovery and restart conditions

Following a fault, the system should not resume operation until the current mechanism positions and locking conditions have been confirmed. Where the mechanical state is known, a controlled return to a reference position may be permitted. Otherwise, operator intervention may be necessary.

Fault clearance alone does not establish readiness for operation. The supervisory controller must verify the relevant positioning, support, locking, and attitude conditions before restarting deployment or recovery.

4.6 Control Evaluation Protocol

4.6.1 Evaluation objectives and operating conditions

The coordinated control framework is considered at two levels: platform attitude regulation and system-level operating coordination. The former concerns levelling accuracy and actuator motion, while the latter concerns operating interlocks, movement completion, and cycle-time constraints.

Representative platform conditions include initial roll and pitch deviations, combined inclination, and time-varying disturbances. However, the original design does not provide the complete geometry, actuator dynamics, or controller parameters required for a validated closed-loop simulation. Any parameterized model must therefore distinguish assumed values from measured equipment characteristics.

4.6.2 Platform-control performance indicators

The principal evaluation indicators are attitude-tracking error, settling time, actuator displacement, and compliance with mechanical limits.

The root-mean-square (RMS) attitude error is defined as

$$e_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left[e_{\phi}^{2}(k) + e_{\theta}^{2}(k)\right]} \tag{45}$$

where, N is the number of samples, and $e_{\phi}(k)$ and $e_{\theta}(k)$ are the roll and pitch errors.

A control strategy should be evaluated not only by its tracking accuracy but also by the actuator movement required to achieve that accuracy. Any comparison between control methods must use consistent operating conditions and actuator constraints.

4.6.3 System-level coordination indicators

System-level evaluation considers the completion of mechanical operations, satisfaction of interlocks, conflicting movement commands, and deployment or recovery duration.

The total operating time depends on individual mechanism durations and the extent of permissible concurrent movement. Although the original design provides nominal roof and centering times, complete cycle-time data are unavailable. Consequently, the operating sequence can be analysed for dependencies, but an actual cycle-time improvement cannot yet be quantified.

Fault scenarios should also be examined to determine whether the supervisory logic prevents prohibited movements and maintains the required mechanical support conditions.

4.6.4 Scope of the numerical evaluation

Section 5 evaluates the mechanical operating requirements using the available engineering parameters, with particular attention to roof loading, centering resistance, and transmission efficiency.

The analysis also considers the coordination constraints governing the docking sequence. However, quantitative claims regarding actual platform settling time, disturbance rejection, or superiority over an existing controller require additional geometric, dynamic, and experimental data.

The evaluation therefore distinguishes reproducible mechanical calculations from control-performance predictions that remain subject to further verification.

5 Numerical Analysis and Performance Evaluation

The analytical models developed in Section 3 were used to evaluate the drive requirements of the sliding roof and centering mechanism under different operating conditions. The analysis examined the effects of friction and transmission efficiency on actuator loading, while the coordination framework in Section 4 was assessed in terms of operating dependencies and mechanical constraints. All numerical results were obtained from the reported engineering parameters and explicitly defined parameter variations rather than physical measurements.

5.1 Numerical Setup and Reference Conditions

The calculations were performed using the original design parameters. For the sliding roof, the moving mass was 125 kg, the nominal friction coefficient was 0.20, and the pinion pitch diameter was 54 mm. The centering mechanism was evaluated using a UAV mass of 130 kg, a friction coefficient of 0.25, a design load factor of 3, and a lead-screw lead of 4 mm.

The sensitivity analysis considered changes in friction and transmission efficiency while maintaining the remaining parameters at their reference values. These variations were introduced to examine the robustness of the nominal drive calculations rather than to represent measured operating conditions.

The reference parameters used in the numerical calculations are listed in Table 6.

Table 6. Reference parameters for numerical analysis

Parameter	Sliding Roof	Centering Mechanism
Moving mass	125 kg	130 kg
Nominal friction coefficient	0.20	0.25
Nominal resistance	250 N	325 N
Design axial load	–	975 N
Transmission parameter	54 mm pinion diameter	4 mm screw lead
Nominal drive torque under stated assumptions	6.75 $\mathrm{N}\cdot\mathrm{m}$	4.51 $\mathrm{N}\cdot\mathrm{m}$
Reported operating time	12.5 s	10 s

Note: "—" means not applicable.

The roof torque refers to the pinion output, whereas the centering torque refers to the motor shaft under an assumed unit transmission ratio. The centering calculation also assumes two equally loaded lead screws and the efficiencies specified in Section 3.3.

5.2 Sliding Roof Performance

The effect of guide friction on the required roof driving torque was examined by varying the effective friction coefficient from 0.10 to 0.35. The panel mass and transmission geometry remained unchanged.

The calculated roof resistance, required driving torque, and nominal torque margin at each friction coefficient are presented in Table 7.

Table 7. Effect of friction coefficient on roof driving torque

Friction Coefficient	Resistance (N)	Required Torque ($\mathrm{N \cdot m}$)	Torque Margin
0.10	125.0	3.38	7.11
0.15	187.5	5.06	4.74
0.20	250.0	6.75	3.56
0.25	312.5	8.44	2.84
0.30	375.0	10.13	2.37
0.35	437.5	11.81	2.03

The required torque increased linearly with the assumed friction coefficient. At the nominal condition, the calculated torque was 6.75 $\mathrm{N \cdot m}$. Increasing the coefficient to 0.30 raised the torque to approximately 10.13 $\mathrm{N \cdot m}$, representing a 50% increase.

The reported gearbox output torque of 24 $\mathrm{N \cdot m}$ exceeded the calculated requirements throughout the examined range. Nevertheless, the available torque margin decreased as friction increased. Additional resistance associated with acceleration, wind loading, and sealing contact was not included in this analysis.

The original design reported a roof opening time of approximately 12.5 s. Based on the specified pinion geometry and an output speed of 60 r/min, the theoretical constant-speed travel time was approximately 11.79 s. The difference reflects the adopted operating-speed assumptions. Actual opening time also depends on acceleration, deceleration, and position confirmation.

5.3 Centering Mechanism Performance

The centering mechanism was evaluated under different friction and transmission-efficiency conditions. The UAV mass and screw geometry were maintained at their nominal values.

5.3.1 Influence of contact resistance

The friction coefficient was varied from 0.15 to 0.35 to examine its influence on the required motor torque. The resulting centering resistance, design axial load, and estimated motor torque are reported in Table 8.

Table 8. Effect of friction coefficient on centering drive requirements

Friction Coefficient	Resistance (N)	Design Axial Load (N)	Estimated Motor Torque at $i_t = 1$ ($\mathrm{N \cdot m}$)
0.15	195	585	2.71
0.20	260	780	3.61
0.25	325	975	4.51
0.30	390	1,170	5.42
0.35	455	1,365	6.32

Note: i_t = motor-to-screw transmission ratio.

Under the assumed unit transmission ratio, the estimated motor torque increased from 2.71 to 6.32 $\mathrm{N \cdot m}$ over the examined friction range. At the nominal friction coefficient of 0.25, the calculated value was approximately 4.51 $\mathrm{N \cdot m}$. Increasing the coefficient to 0.35 raised this estimate by approximately 40%.

These results indicate that contact conditions can substantially affect the centering drive requirements. The estimates remain dependent on the assumed load distribution between the two lead screws.

5.3.2 Influence of transmission efficiency

The lead-screw efficiency was varied while maintaining the nominal axial load, an additional transmission efficiency of 0.50, and a unit motor-to-screw speed ratio. The corresponding motor torque estimates for the different lead-screw efficiencies are given in Table 9.

Table 9. Effect of lead-screw efficiency on motor torque

Lead-Screw Efficiency	Estimated Motor Torque at $i_t = 1$ ($\mathrm{N} \cdot \mathrm{m}$)
0.35	7.09
0.40	6.21
0.45	5.52
0.50	4.97
0.55	4.51
0.60	4.14

Note: i_t = motor-to-screw transmission ratio.

The required torque decreased as the assumed lead-screw efficiency increased. Reducing efficiency from 0.55 to 0.35 increased the estimated motor torque from 4.51 to 7.09 $\mathrm{N} \cdot \mathrm{m}$.

The sensitivity results show that the assumed lead-screw efficiency has a substantial effect on the estimated motor torque. However, the efficiency values used in this analysis do not establish the self-locking capability of the transmission. That requirement must be checked separately using the actual thread geometry and friction conditions, together with the holding capacity of the mechanical locking elements.

At the reported motor speed of 1,200 r/min, the nominal torque of 4.51 $\mathrm{N} \cdot \mathrm{m}$ corresponded to approximately 0.57 kW of mechanical shaft power. The reported 10 s centering time could not be independently verified because the complete transmission ratio and centering stroke were unavailable.

5.4 Closed-Loop Evaluation of the Three-Degree-of-Freedom Landing Platform

A numerical study was conducted to examine the feasibility of closed-loop attitude regulation using a representative three-actuator landing platform. Since the geometric and dynamic parameters of the manufactured docking platform were unavailable, a simplified small-angle kinematic model was adopted. The three actuators were positioned at equal angular intervals on a circle with a radius of 0.30 m. Their relative displacements were limited to $\pm$50 mm, and the maximum actuator velocity was set to 40 mm/s. A first-order actuator response with a time constant of 0.20 s was assumed.

$$\tau_{\mathrm{a}} \dot{\Delta l}_i(t) + \Delta l_i(t) = \Delta l_{i,\mathrm{cmd}}(t) \tag{46}$$

where, τ_{a} = 0.20 s is the assumed actuator time constant, Δl_i is the actual relative actuator displacement, and $\Delta l_{i,\mathrm{cmd}}$ is the commanded displacement. The actuator motion was also subject to the prescribed travel and velocity limits.

The attachment coordinates of the three actuators were defined as

$$\begin{aligned} x_i &= R\cos\alpha_i, \\ y_i &= R\sin\alpha_i, \end{aligned} \qquad i = 1, 2, 3 \tag{47}$$

where, R = 0.30 m is the radial distance from the platform centre to each actuator attachment point, and $\alpha_i \in \{90^\circ, 210^\circ, 330^\circ\}$ denotes the angular position of the corresponding actuator.

The general inverse kinematic formulation in Section 3.2 was reduced to a small-angle representation with fixed nominal platform height. Under this assumption, the relative vertical displacement at each actuator attachment point was approximated as

$$\Delta l_i = h + \phi y_i - \theta x_i \tag{48}$$

where, h denotes the common vertical displacement, and ϕ and θ are the platform roll and pitch angles in radians. For the numerical evaluation, the nominal height was held constant, and $h = 0$ was adopted in the relative displacement calculation. This linearized relationship was used to generate coordinated actuator commands. A common motion-scaling procedure was applied when necessary to preserve the relative actuator movements while respecting the prescribed displacement and velocity limits. The approximation assumes that the actuator displacements are measured along the nominal vertical direction and that the platform rotations remain sufficiently small for higher-order terms to be neglected. It does not represent the full nonlinear kinematics of the physical mechanism.

P and PD feedback controllers were evaluated under static inclination and periodic disturbance conditions. The P gain was set to 0.85 for both controllers, and the derivative gain was 0.055 s for PD control. The derivative term

was filtered using a first-order discrete filter with coefficients of 0.90 and 0.10 for the previous filtered value and the current error derivative, respectively. The attitude correction command was limited to $\pm 10^\circ$ on each axis.

The discrete feedback calculation was represented by

$$\begin{aligned} e[k] &= \eta_{\mathrm{d}}[k] - \eta_{\mathrm{m}}[k] \\ d[k] &= \frac{e[k] - e[k-1]}{\Delta t} \\ d_{\mathrm{f}}[k] &= 0.90 d_{\mathrm{f}}[k-1] + 0.10 d[k] \\ u[k] &= 0.85 e[k] + 0.055 d_{\mathrm{f}}[k] \end{aligned} \tag{49}$$

where, Δt = 0.01 s is the sampling interval, $d_{\mathrm{f}}[k]$ is the filtered derivative of the attitude error, and $u[k]$ is the feedback correction before saturation. For P control, the derivative term was omitted. The feedback output was added to the current relative platform attitude to generate the next attitude command, as expressed in Eq. (50).

$$\eta_{\mathrm{c}}[k] = \mathrm{sat}_{\pm 10^\circ}\left(\eta_{\mathrm{r}}[k] + u[k]\right) \tag{50}$$

where, $\eta_{\mathrm{c}}[k]$ is the updated platform attitude command, $\eta_{\mathrm{r}}[k]$ is the current relative platform attitude calculated from the actuator displacements, and $u[k]$ is the feedback correction obtained from Eq. (49). The operator $\mathrm{sat}_{\pm 10^\circ}$ limits each component of the commanded attitude to the range of -10° to 10°. The updated attitude command was then converted into actuator displacements using the linearized kinematic relationship in Eq. (48). A common scaling factor was applied to the actuator increments when necessary to satisfy the prescribed travel and velocity limits.

The simulations were conducted over 12 s with a numerical time step of 0.01 s. For the periodic disturbance test, the roll and pitch amplitudes were 3° and 2°, respectively, at a frequency of 0.4 Hz, with a pitch phase offset of 0.6 rad relative to roll. The washout-based reference-generation arrangement discussed in Section 4.2.2 was not included in the numerical comparison, which was limited to P and PD attitude feedback.

All geometric parameters, actuator characteristics, and controller settings were assumed for the numerical evaluation rather than obtained from the manufactured docking system. The resulting control responses therefore describe the behaviour of the representative model under the specified conditions and should not be interpreted as experimentally verified performance of the physical docking platform. These parameters were selected to define a representative numerical test case rather than to reproduce a specific commercial actuator or manufactured platform. Their influence on the predicted control response should be examined before the model is used for equipment-level design decisions.

5.4.1 Static levelling response

The initial platform roll and pitch angles were set to 5° and -4°, respectively. Both feedback strategies reduced the inclination toward the horizontal reference. Settling time was defined as the earliest time after which both roll and pitch errors remained within $\pm 0.2^\circ$ for the remainder of the static simulation. Under this criterion, the P and PD controllers achieved settling times of 1.02 s and 1.21 s, respectively. The simulated roll and pitch responses under the two feedback strategies are compared in Figure 8. Both controllers reduced the initial inclination, although P control reached the prescribed settling tolerance earlier under the selected conditions.

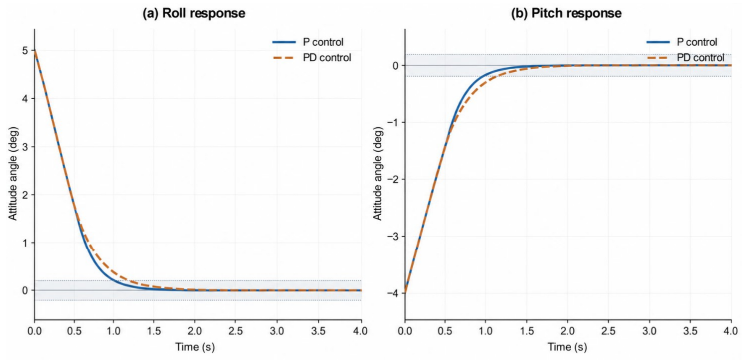


Figure 8. Static roll and pitch responses of the representative three-actuator platform under P and PD control.
Note: P = proportional; PD = proportional-derivative.

The PD controller therefore did not provide a shorter settling time under the selected conditions. This difference reflects the interaction between controller parameters and actuator dynamics and should not be interpreted as a general ranking of the two strategies.

5.4.2 Response to periodic disturbances

Periodic base-inclination disturbances were applied to the roll and pitch axes, with amplitudes of 3° and 2°, respectively, at a frequency of 0.4 Hz. The pitch disturbance was assigned a phase offset of 0.6 rad relative to roll. The combined RMS attitude error was calculated over the interval from 4 to 12 s.

The uncompensated model produced an RMS error of 2.599°, compared with 1.299° under P feedback and 1.185° under PD feedback. The PD controller reduced the RMS error by approximately 54.4% relative to the uncompensated case and by 8.8% relative to P feedback. Figure 9 compares the simulated attitude responses under periodic disturbances for the uncompensated, P-controlled, and PD-controlled cases. The feedback strategies reduced the residual attitude variation, with PD control producing the lowest combined RMS error under the assumed conditions.

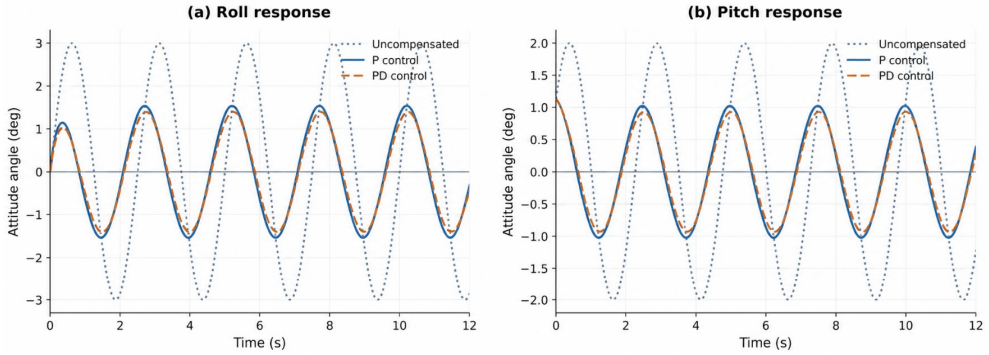


Figure 9. Attitude regulation under periodic disturbances: comparison of uncompensated, P-controlled, and PD-controlled responses

Note: P = proportional; PD = proportional-derivative.

These results indicate that closed-loop compensation reduced the attitude error under the assumed disturbance conditions. The relative performance of the two controllers remains dependent on their gains and the simplified actuator model.

5.4.3 Actuator coordination and motion constraints

The actuator commands were generated using the linearized kinematic relationship in Eq. (48), with a common motion-scaling procedure applied to maintain the prescribed displacement and velocity limits.

Under periodic disturbance with PD feedback, the maximum absolute actuator displacement was 13.55 mm, and the maximum actuator velocity was 33.61 mm/s. Both values remained within the assumed travel and velocity limits of ±50 mm and 40 mm/s. The displacement histories of the three actuators during the periodic disturbance test are presented in Figure 10. The coordinated movements remained within the assumed travel limits, while the corresponding maximum actuator velocity was below the prescribed limit of 40 mm/s.

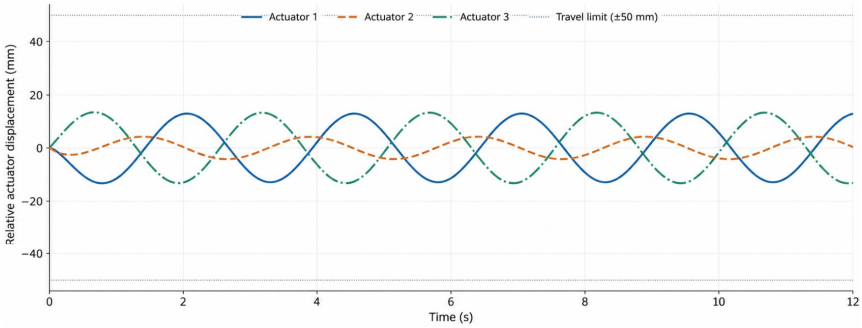


Figure 10. Coordinated actuator displacements of the representative three-actuator platform under periodic disturbances with PD control

Note: PD = proportional-derivative.

The common motion-scaling procedure maintained the prescribed relative actuator displacements in the representative kinematic model. This result does not establish the mechanical feasibility of the actual actuator and joint configuration.

The main performance results of the three control cases are summarized in Table 10. Both feedback controllers reduced the attitude error under periodic disturbances, with PD control producing the lowest RMS error. However, P control achieved a shorter settling time during static levelling. The actuator displacement and velocity recorded in the dynamic PD case also remained within the prescribed limits.

Table 10. Closed-loop performance of the representative three-actuator platform

Performance Indicator	No Feedback	P Control	PD Control
Static settling time (s)	–	1.02	1.21
Dynamic RMS attitude error (°)	2.599	1.299	1.185
RMS error reduction relative to no feedback (%)	–	50.0	54.4
Maximum actuator displacement under dynamic PD control (mm)	–	–	13.55
Maximum actuator velocity under dynamic PD control (mm/s)	–	–	33.61

Note: P = proportional; PD = proportional-derivative; RMS = root-mean-square; "–" means not applicable. All results were obtained using an assumed platform geometry and actuator model. The displacement and velocity results are reported only for the dynamic PD case.

5.4.4 Scope and limitations of the simulation

The numerical model was intended to assess control feasibility under explicitly assumed conditions. It did not include the actual attachment-point geometry, joint constraints, actuator force limits, structural deformation, sensor noise, or a calibrated dynamic model of the manufactured platform.

Consequently, the reported settling times, RMS errors, and actuator movements are model-based predictions rather than measured performance values. Their applicability to the physical docking station requires further geometric identification, dynamic modelling, and experimental verification.

5.5 Coordinated Operation and Control Constraints

The deployment and recovery procedures involve both sequential and potentially overlapping movements. Roof opening and early-stage lifting may proceed concurrently when sufficient clearance is available, whereas aircraft centering and locking must follow touchdown confirmation.

The principal operating dependencies are summarized in Table 11.

Table 11. Operating dependencies of the docking system

Operation	Required Condition	Coordination Requirement
Parking platform extraction	Transfer mechanism engaged	Storage lock released only after engagement
Roof opening and lifting	Sufficient mechanical clearance	Concurrent movement permitted within clearance limits
Platform levelling	Platform at operating position	Levelling completed before UAV authorization
Aircraft centering	Touchdown confirmed	Centering initiated after landing
Internal transfer	Aircraft positioned and locked	Transfer inhibited until locking confirmation
Service reconnection	Platform secured in storage	Connectors actuated after positioning

Note: UAV = unmanned aerial vehicle.

The available operating times cover the roof and centering mechanisms but not the complete deployment and recovery sequences. The analysis therefore identifies permissible concurrent operations and mandatory interlocks without assigning a numerical improvement in cycle time. A quantitative comparison requires the remaining actuator travel times and the corresponding clearance conditions.

The 3-DOF platform was also considered in terms of its kinematic and control constraints. The inverse kinematic model established in Section 3.2 provides a basis for calculating actuator displacements, while the control framework in Section 4 defines attitude-tracking and operating-authorization requirements. The closed-loop results reported in Section 5.4 were obtained using a representative three-actuator model with assumed geometric and dynamic parameters. They provide a numerical assessment of attitude regulation under the selected conditions but do not constitute validation of the manufactured platform.

5.6 Discussion of Numerical Results

The numerical analysis reproduced the nominal mechanical requirements of the original docking-station design and identified their sensitivity to operating assumptions. For the sliding roof, the calculated driving torque increased from 3.38 to 11.81 $\mathrm{N \cdot m}$ as the assumed friction coefficient increased from 0.10 to 0.35. The reported gearbox output torque exceeded these values, although the nominal torque margin decreased substantially at higher friction levels.

For the centering mechanism, the required motor torque was strongly influenced by both contact resistance and lead-screw efficiency. Under the nominal assumptions, the calculated torque was approximately 4.51 $\mathrm{N \cdot m}$. Increasing the friction coefficient or reducing transmission efficiency raised the drive requirement, indicating that the original motor selection should be checked against the expected operating range rather than a single nominal condition.

The representative closed-loop simulation provided a separate assessment of platform levelling. Both P and PD feedback reduced the combined attitude error under periodic disturbances, although P control settled more quickly in the static test. The actuator movements remained within the assumed limits. These results support the feasibility of feedback-based levelling in the simplified model, but they do not establish the performance of the physical docking platform.

The coordination analysis identified the principal mechanical dependencies governing unattended deployment and recovery. It also showed that some operations may overlap without violating the required sequence, provided that clearance and locking conditions are satisfied. However, the available data did not support a quantitative estimate of complete cycle time or control-related efficiency improvement.

Overall, the results establish a reproducible basis for evaluating mechanical loading and operating constraints. They also identify the parameters requiring further verification, particularly the actual transmission characteristics and the dynamic response of the 3-DOF landing platform. These limitations should be considered when assessing the reliability of the complete docking system.

6 Discussion

6.1 Mechanical Design and Its Implications for Automated Operation

The sensitivity analysis indicates that drive selection for unattended docking equipment should account for variations in mechanical resistance rather than nominal loading alone. Guide friction, landing-gear contact conditions, and transmission losses may change during repeated operation and reduce the available torque margin. These effects also have implications for supervisory control, since delayed or incomplete movements must be detected before subsequent operations are authorized.

The results also highlight the relationship between mechanical design and supervisory control. Variations in load may affect the time required for a mechanism to reach its designated position or may prevent completion when the available drive capacity is insufficient. Position confirmation, timeout monitoring, and operating interlocks are consequently necessary complements to mechanical drive selection. The calculations do not establish long-term reliability, but they identify conditions that should be considered during equipment commissioning and subsequent verification.

6.2 Role of the Three-Degree-of-Freedom Platform in Unattended Docking

The 3-DOF landing platform distinguishes the investigated docking station from arrangements that rely entirely on a fixed landing surface. Its intended function is to compensate for inclination of the supporting structure and establish the required platform attitude before aircraft take-off or landing. This capability is particularly relevant to installations where the docking enclosure may be mounted on an inclined foundation or a moving support.

The inverse kinematic model established the relationship between platform attitude and actuator displacement, while the numerical study examined closed-loop regulation using a representative three-actuator arrangement. Under the assumed conditions, both P and PD feedback reduced the attitude error. PD control produced a lower RMS error during periodic disturbance, whereas P control achieved a shorter settling time during static levelling. The comparison indicates that the choice of controller depends on the operating requirement rather than a single performance indicator.

These results were obtained using assumed geometry, actuator dynamics, and controller parameters. Although the model satisfied its prescribed displacement and velocity constraints, the physical platform has not been verified against these assumptions. In particular, actuator force capacity, joint articulation, structural deformation, and sensor characteristics remain outside the present numerical model.

This limitation is important when considering industrial deployment. A platform capable of correcting a static inclination may not necessarily provide adequate compensation under rapid support motion. Its practical operating

range depends on actuator speed, available stroke, control bandwidth, and the magnitude and frequency of the disturbance. These factors should be examined together in subsequent numerical and experimental studies.

6.3 Coordination and System-Level Operating Requirements

The docking station combines several mechanical operations within a shared working space. Roof movement, platform lifting, horizontal transfer, attitude adjustment, centering, and locking must therefore be coordinated according to the current operating conditions.

The control framework developed in Section 4 formalizes these relationships through discrete operating states and supervisory interlocks. In particular, the storage lock must remain engaged until the transfer mechanism has established the required support, while the lifting platform must not enter the roof-opening region without sufficient clearance. During recovery, centering and locking must be confirmed before internal transfer is authorized.

These conditions illustrate that coordinated operation is not simply a matter of executing a predetermined sequence. The controller must also verify the mechanical state reached after each movement. This distinction is important because an actuator command may be completed without the associated positioning or locking requirement being satisfied.

The analysis further indicates that certain movements may overlap when their mechanical constraints permit. Roof opening and early-stage lifting provide one example. Nevertheless, the available operating-time data were insufficient to quantify any reduction in the complete deployment or recovery cycle. The contribution of the present framework is therefore the explicit organization of operating dependencies rather than a demonstrated improvement in cycle-time efficiency.

From an industrial automation perspective, the integration of continuous platform regulation with discrete mechanical coordination provides a useful basis for developing unattended docking equipment. Its practical effectiveness, however, depends on the implementation and verification of the corresponding control logic, sensors, and protective functions.

6.4 Limitations and Further Verification

The numerical analysis was based on the reported engineering parameters and assumed variations in friction and transmission efficiency. Although the calculations are reproducible, they do not account for acceleration, mechanical misalignment, environmental loading, or unequal force distribution. The self-locking behaviour of the centering transmission also requires verification using the actual screw geometry and friction conditions.

The closed-loop simulation was conducted using a representative three-actuator model rather than the actual docking-platform geometry. Although the assumed kinematic constraints were satisfied, the model did not account for actuator force limits, structural flexibility, joint constraints, or sensor uncertainty. The predicted control performance must therefore be verified using the actual mechanical configuration and measured actuator characteristics. Further testing should also examine controller robustness under different disturbance frequencies and operating loads.

System-level verification is also required. Repeated deployment and recovery tests should examine operating-cycle duration, interlock behaviour, fault recovery, and the reliability of mechanical positioning and locking. Particular attention should be given to the interaction between roof movement and lifting, the transfer of mechanical support between locking mechanisms, and the completion of aircraft restraint before storage.

These investigations are necessary to establish whether the proposed mechanical arrangement and coordinated control framework can satisfy the reliability requirements of unattended industrial operation.

7 Conclusions

This study investigated the mechanical design and coordinated operation of a four-position automated UAV docking system intended for unattended industrial deployment and recovery. The original engineering configuration was retained, including the sliding roof, 3-DOF landing platform, and mechanical centering and locking assembly. Analytical modelling and numerical sensitivity analysis were used to examine the principal drive requirements, while a supervisory control framework was formulated to describe the coordination of mechanical operations.

The main findings are summarized as follows.

(1) The sliding roof required a nominal driving torque of 6.75 $\mathrm{N}\cdot\mathrm{m}$ under the original design assumptions. Increasing the effective friction coefficient from 0.10 to 0.35 raised the calculated torque from 3.38 to 11.81 $\mathrm{N}\cdot\mathrm{m}$. The reported gearbox output torque of 24 $\mathrm{N}\cdot\mathrm{m}$ exceeded these calculated requirements, although the nominal torque margin decreased at higher friction levels.

(2) Under the assumed unit transmission ratio, the centering mechanism required an estimated motor torque of 4.51 $\mathrm{N}\cdot\mathrm{m}$ at the reference condition. The calculated value increased to 6.32 $\mathrm{N}\cdot\mathrm{m}$ when the contact friction coefficient reached 0.35 and to 7.09 $\mathrm{N}\cdot\mathrm{m}$ when the assumed lead-screw efficiency decreased to 0.35. These results identified contact resistance and transmission efficiency as important factors affecting the estimated drive requirements.

(3) A representative three-actuator model was used to examine closed-loop platform levelling under assumed operating conditions. The P and PD controllers achieved static settling times of 1.02 s and 1.21 s, respectively. Under periodic disturbances, the combined RMS attitude error decreased from 2.599° without feedback to 1.299° with P control and 1.185° with PD control. The maximum actuator displacement and velocity in the dynamic PD case remained within the prescribed limits. These results demonstrate the feasibility of attitude compensation in the simplified numerical model, rather than the verified performance of the physical docking platform.

(4) The supervisory control framework defined the operating dependencies and interlocking conditions for roof movement, lifting, platform levelling, aircraft centering, locking, and internal transfer. It also identified operations that may overlap within the mechanical constraints. Quantitative verification of the complete operating cycle and control performance remains necessary.

The findings provide an integrated basis for assessing the mechanical and control requirements of unattended UAV docking equipment. The drive calculations and sensitivity analyses identify the principal loading conditions, while the supervisory framework defines the operating dependencies between the docking mechanisms. The closed-loop numerical study further illustrates the potential of coordinated actuator control for platform levelling. Validation using the actual platform geometry, actuator characteristics, and experimental measurements remains necessary before the predicted performance can be applied to the manufactured system.

Author Contributions

Conceptualization, X.K.Z. and Y.X.T.; methodology, X.K.Z.; validation, X.K.Z.; formal analysis, X.K.Z.; investigation, X.K.Z.; data curation, X.K.Z.; writing—original draft preparation, X.K.Z.; writing—review and editing, Y.X.T.; visualization, Y.X.T.; project administration, X.K.Z.; funding acquisition, X.K.Z. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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